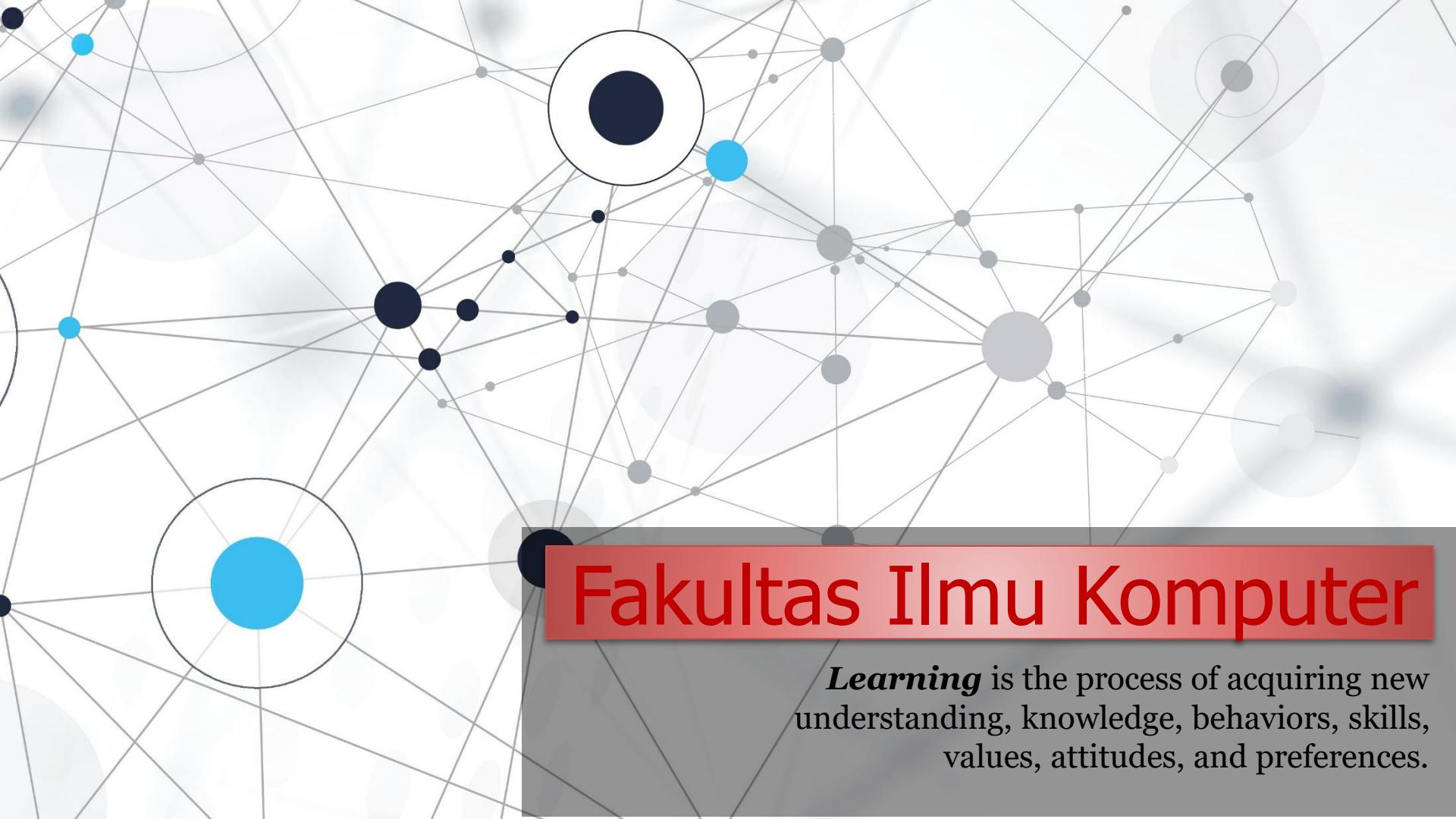


An abstract network diagram with various sized nodes (black, blue, grey) connected by thin grey lines. Some nodes are highlighted with larger circles. The background is white with faint grey circles.

Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

Introduction to **DevOps Engineering**



**Welcome to *the* DevOps
Engineering course**

A decorative border of green leaves and branches at the top of the slide.

1 - What is Development Operation Engineering ?

A decorative border of green leaves and branches at the bottom of the slide.

- 1) Pengenalan konsep DevOps
- 2) Perbedaan antara DevOps dan model pengembangan tradisional
- 3) Prinsip-prinsip dan praktik DevOps
- 4) Tool dan teknologi yang digunakan dalam DevOps



Dalam pengembangan sebuah sistem, peran seorang DevOps Engineer dalam lingkungan server melibatkan berbagai tugas dan tanggung jawab yang bertujuan untuk mengintegrasikan pengembangan (*development*) dan operasi (*operations*) guna meningkatkan efisiensi, keamanan dan reliabilitas proses pengembangan dan pengelolaan infrastruktur.

DevOps Engineer memiliki peranan yang sangat penting dalam pengembangan dan pengelolaan aplikasi perangkat lunak untuk mengintegrasikan proses *development* dengan *operations*, sehingga memungkinkan pengembangan aplikasi yang lebih *cepat*, *handal* dan *aman*.



Development & Operation

Full Stack Web Development

merujuk pada kemampuan seseorang untuk mengembangkan dan mengelola aplikasi web dari *sisi front-end* hingga *sisi back-end*. Seorang Full Stack Web Developer memiliki pengetahuan dan keterampilan dalam berbagai teknologi dan bahasa pemrograman yang digunakan untuk mengembangkan antarmuka pengguna (UI/UX) serta **mengelola server**, *database*, dan *logika bisnis* di sisi *server*.

Seorang **DevOps Engineer** adalah profesional yang bertanggung jawab untuk memastikan **pengiriman software yang cepat, andal dan berkualitas tinggi**. DevOps Engineer menggabungkan praktik-praktik dari pengembangan perangkat lunak (*Development*) dengan operasional TI (*Operations*) untuk menciptakan aliran kerja (*workflow*) yang efisien dan ter-automasi dalam proses pengembangan, pengujian, dan penyebaran aplikasi.

Hubungan antara Full Stack Web Development dan DevOps Engineer terletak dalam fokus yang sama pada aspek-aspek pengembangan *software/app/website*. Seorang *full stack web developer* yang berpengalaman dapat menggabungkan pengetahuan dalam mengembangkan aplikasi web dengan pemahaman tentang **praktik DevOps** untuk memastikan *deploy* aplikasi yang **sukses** dan **efisien**.

Waterfall vs Agile vs DevOps

	 Waterfall	 Agile	 DevOps
Approach	Waterfall model provides a linear sequential approach to managing software projects. Each phase depends on deliverables from the previous one.	Agile is a highly dynamic and iterative approach where you do not need the complete set of requirements to start with. You can develop some features and check customer response before next steps	DevOps is an Agile methodology encompassing Development (Dev) and Operations (Ops). It enables end-to-end lifecycle delivery of features, fixes, and updates at frequent intervals.
Year	1970	2001	2009
Scope and Schedule	Adjust schedule to preserve scope. Fixed requirements. Limited flexibility.	Adjust scope to preserve schedule. Iterative approach allows for prioritization	Adjust scope to preserve schedule. Highly responsive to business needs
Time to Market	Slow	Rapid	Continuous
Automation	Low	Varied	High
Customer Feedback	End of Project	After every sprint	Continuous
Quality	Low. Issues are identified only at testing stage	Better. Issues are identified after every sprint	High. Automated unit testing during development
Collaboration	Teams operate in silos	Multiple teams are involved, but not all	All stakeholders are involved from start to finish
Variations	V-model	Scrum, XP, LeSS, SAFe	CI/CD, DevSecOps

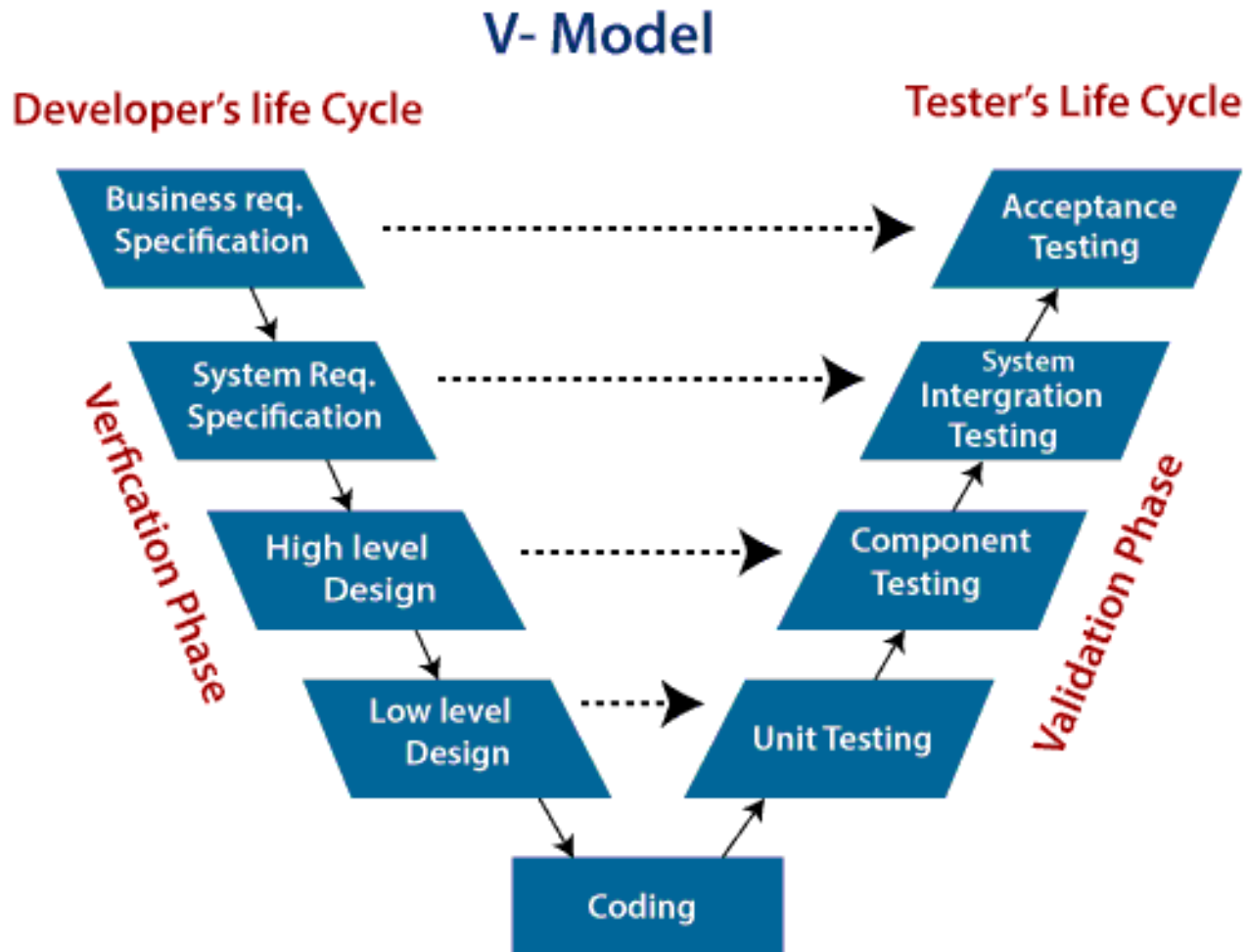
Team Operate *in Silos*

“Teams operate in silos” means that teams work separately or in isolation from each other, **without communication or collaboration** with other teams. Each team or department only focuses on their own tasks or goals, without sharing information, strategies, or ideas with other teams in the organization.



V-Model *vs* CI/CD

V-Model (Verification and Validation Model) is one of the software development methodologies that follows a sequential approach (like Waterfall), but with a strong emphasis on testing at every stage of development.



V-Model vs CI/CD

CI/CD is one of the DevOps practices used to make software development more organized. The use of CI/CD in application development provides many benefits that will be immediately felt by the development team. In addition, there is a large selection of tools that can be used to support the running of this CI / CD process automatically and safely.



Compare DevOps to Traditional IT

<u>Dimensions</u>		<u>Traditional IT</u>	<u>DevOps</u>
Planning & Organization	Batch Size	Big	Micro
	Organization	Skill Centric Silos	Dedicated Cells
	Scheduling	Centralized	Decentralized & Continuous
Performance & Culture	Release	High Risk Event	Non Event
	Information	Disseminated	Actionable
	Culture	Do Not Fail	Fail Early
Measure	Metric	Cost & Capacity	Cost, Capacity, and Flow (Time)
	Define "Done"	<i>"I did my job"</i>	<i>"Its ready to deploy"</i>

From Skill Centric Silos to Dedicated Cells

Traditional IT is organized around skill centric silos. For the most part silos work, they band like skills together, drive greater utilization, and benefit from economies of scale. However, where these “cost optimized” silos break down is at the hand offs. In a typical IT environment, a new feature has to go through at least 3 – 4 silos before the customer gets it. It is not uncommon for an idea/code to spend 80% of its time waiting or ping-ponging back and forth between silos.

A DevOps organization also operates in a silos but at a different cross section. Here, teams are arranged in “cells”, consisting of dedicated cross functional teams, and focused on only one application. By creating this self-sufficient cell that consists of developers, testers, business analysts, and operators, an idea can move from one stage to another without hand offs, promotes cross-training/understanding (key to collaboration), focuses the team on the end goal, and encourages the “*shift left*” thinking. As for utilization and scale benefits, the rules for response states that for every $\frac{1}{4}$ reduction in cycle time, productivity will improve 2x, and operating costs by 20%.

Jika kalian menjadi Seorang
DevOps Engineer bagian penting
apa yang harus di perhatikan?

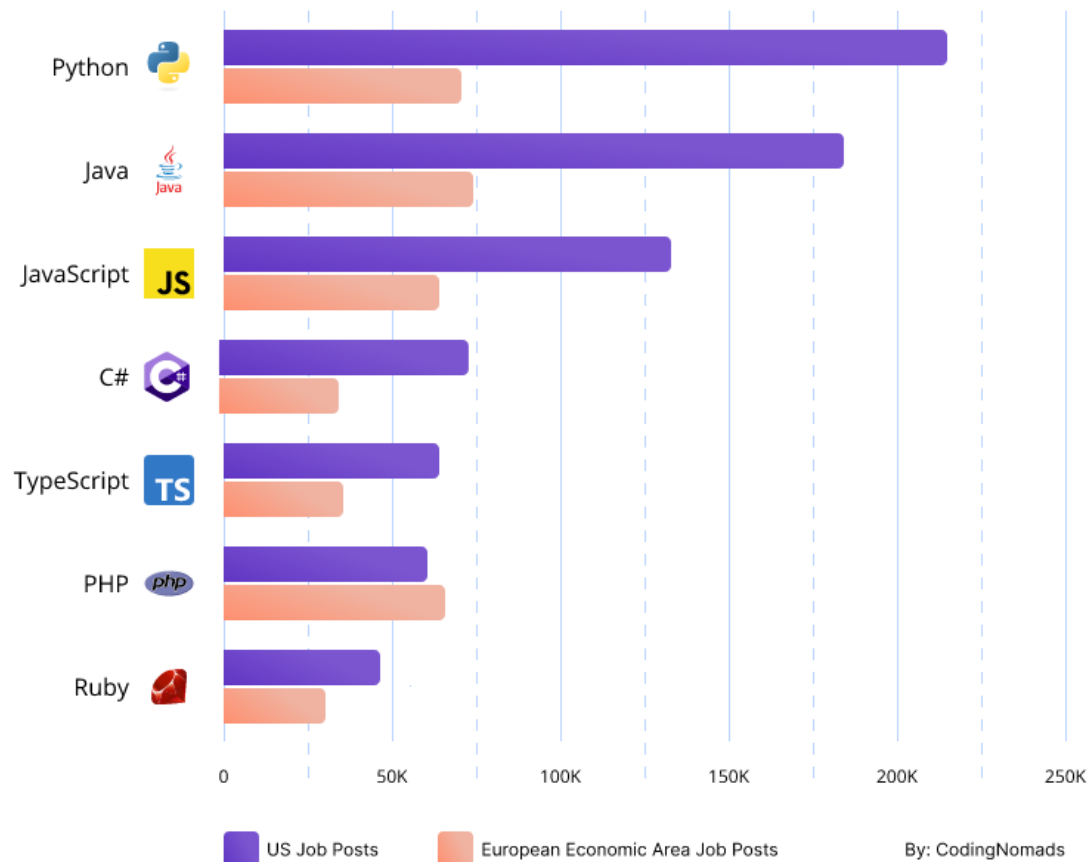
DevOps Engineering Approach

Automatisasi	a) Otomatisasi proses pengembangan, pengujian dan deploy aplikasi untuk mempercepat dan mengoptimalkan alur kerja. b) Otomatisasi manajemen infrastruktur dan konfigurasi untuk menjaga konsistensi dan menghindari human error.
Keamanan	a) Mengamankan aplikasi web dengan konfigurasi keamanan yang tepat, seperti manajemen akses dan otentikasi yang kuat. b) Monitoring lalu lintas jaringan dan aktivitas mencurigakan untuk mendeteksi dan merespons serangan potensial.
Penyediaan Infrastruktur	a) Manage infrastruktur cloud atau on-premise yang dibutuhkan untuk menjalankan aplikasi web. b) Menggunakan teknologi kontainer dan orkestrasi seperti Docker dan Kubernetes untuk skalabilitas dan pengelolaan aplikasi yang lebih baik.
Monitoring dan Pemantauan	a) Apply sistem pemantauan yang efektif untuk memantau kinerja, ketersediaan, dan masalah aplikasi web. b) Menggunakan alat pemantauan untuk mendapatkan wawasan yang mendalam tentang infrastruktur dan aplikasi.
CI/CD	a) Apply praktik kontinu integrasi (CI) dan kontinu pengiriman (CD) untuk memastikan pengiriman perubahan kode yang cepat dan berkualitas tinggi. b) Membangun dan memelihara alur kerja otomatis untuk pengujian dan penyebaran perubahan kode.
Kolaborasi Tim	a) Bekerja sama dengan tim pengembang, operasional TI dan anggota tim lainnya untuk memastikan alur kerja yang efisien dan kolaborasi yang baik. b) Berbagi pengetahuan dan pengalaman dengan tim untuk meningkatkan pemahaman dan keterampilan dalam pengembangan dan pengelolaan aplikasi web.

Top Programming Languages

Most In-Demand Programing Languages of 2022

Based on LinkedIn Job Posting in the USA & Europe



<https://devtorium.com/blog/top-7-programming-languages-for-web-design-and-development-services/>

Most Popular Web Frameworks

Mengapa harus menggunakan *framework*?



Most popular web frameworks



stackoverflow.com - 2021 Developer Survey



Modern Web Development

Modern web development is a term that refers to the process of creating a **website** or **web application** that takes advantage of **the latest web technologies**. This can include using a modern web development framework, such as **AngularJS** or **React**, or using modern tools and services, such as **Cloud Service**, **GitHub** or **Heroku**.

What is Web Developer?

Web developer adalah sebuah profesi dimana ini adalah orang yang bertugas membuat dan mengembangkan *website*, atau istilah lainnya *web development*.



Back-end web developer	Web developer yang tugasnya berkaitan dengan <i>server</i> . Dimana developer ini bertanggung jawab untuk memastikan agar website selalu cepat (<i>fast</i>), aman (<i>secure</i>), dan berfungsi sesuai tujuan penggunaan (<i>user</i>).
Front-end web developer	Web developer ini bertugas untuk mendesain antarmuka website. Tujuannya, agar <i>website</i> memiliki tampilan yang baik dan fitur-fitur yang mudah digunakan pengunjung (<i>end user</i>).
Full-stack web developer	Web developer yang memiliki dua tugas sekaligus, baik melakukan pekerjaan <i>front end</i> dan maupun <i>back end</i> . Full-stack developer harus mampu membuat tampilan dan fitur website, sembari mengoptimalkan kecepatan dan keamanan dari sisi <i>server</i> .
DevOps Engineer	Seorang engineer yang bertanggung jawab untuk mengelola dan memelihara infrastruktur serta aliran kerja (<i>workflow</i>) pengembangan dan pengelolaan aplikasi web secara efisien.

What is LAMP Stack?



LAMP stack is a popular open-source web development platform. LAMP Stack development is popular web development methodology that allow web developers to build projects quickly, efficiently, and cost-effectively.

The LAMP Stack is a group of open-source software technologies that work together to run dynamic websites and applications. LAMP, which stands for Linux, Apache, MySQL, and PHP, allows you to build websites quickly and easily without buying expensive third-party software licenses.

LAMP Stack Alternatives:

- ✓ **MEAN** stack (MongoDB, Express, Angular, Node. js)
- ✓ **LEMP** stack (Linux, NGINX, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **LAPP** stack (Linux, Apache, PostgreSQL, PHP)
- ✓ **LEAP** stack (Linux, Eucalyptus, AppScale, Python)
- ✓ **LLMP** stack (Linux, Lighttpd, MySQL/MariaDB, PHP/Perl/Python)
- ✓ **XAMPP** stack (Cross-platform, Apache, MariaDB, PHP, Perl)

Menurut anda, apakah seorang
DevOps Engineer adalah
seorang web developer?



Seorang DevOps Engineer

Seorang web developer akan lebih terlibat dalam pembuatan dan pengkodean aplikasi web, sedangkan seorang DevOps Engineer berfokus pada infrastruktur dan alur kerja untuk menjalankan dan mengelola sistem tersebut.

- ❖ Seorang DevOps Engineer di web tidak selalu menjadi seorang web developer, meskipun pengetahuan dan keterampilan dalam pengembangan web sering kali merupakan aset yang berharga dalam peran tersebut.
- ❖ DevOps Engineer berfokus pada pengembangan dan pengelolaan infrastruktur dan aliran kerja (workflow) yang mendukung pengembangan dan pengelolaan aplikasi web secara efisien. Mereka memastikan bahwa alur kerja pengembangan, pengujian, dan penyebaran aplikasi berjalan lancar, serta mengoptimalkan kinerja dan ketersediaan infrastruktur yang mendukung aplikasi web tersebut.

Full Stack Web Developer

A *full stack web developer* is a person who can develop both client and server software. In addition to mastering HTML and CSS, we also need to know how to:

Program a browser

(like using JavaScript, jQuery, Angular, or Vue)

Program a server

(like using PHP, ASP, Python, or Node)

Program a database

(like using MySQL, SQLite, or MongoDB)

Front End & Back End

Popular Stacks

LAMP stack: JavaScript - Linux - Apache - MySQL - PHP

LEMP stack: JavaScript - Linux - Nginx - MySQL - PHP

MEAN stack: JavaScript - MongoDB - Express - AngularJS - Node.js

Django stack: JavaScript - Python - Django - MySQL

Ruby on Rails: JavaScript - Ruby - SQLite - Rails

Client Software (Front End)

- [HTML](#)
- [CSS](#)
- [Bootstrap](#)
- [W3.CSS](#)
- [JavaScript](#)
- [ES5](#)
- [HTML DOM](#)
- [JSON](#)
- [XML](#)
- [jQuery](#)
- [Angular](#)
- [React](#)
- [Backbone.js](#)
- [Ember.js](#)
- [Redux](#)
- [Storybook](#)
- [GraphQL](#)
- [Meteor.js](#)
- [Grunt](#)
- [Gulp](#)

Server Software (Back End)

- [PHP](#)
- [ASP](#)
- [C++](#)
- [C#](#)
- [Java](#)
- [Python](#)
- [Node.js](#)
- [Express.js](#)
- [Ruby](#)
- [REST](#)
- [Go](#)
- [SQL](#)
- [MongoDB](#)
- [Sass](#)
- [Less](#)
- [Firebase.com](#)
- [Parse.com](#)
- [PaaS \(Azure and Heroku\)](#)

Advantages

The advantage of being a **full stack web developer** is:

- 1) You can **master all the techniques** involved in a development project.
- 2) You can make a prototype very rapidly.
- 3) You can **provide help to all the team members**.
- 4) You can **reduce the cost** of the project.
- 5) You can **reduce the time used** for team communication.
- 6) You **can switch between front and back end development** based on requirements.
- 7) You can **better understand all aspects** of new and upcoming technologies.

Disadvantages

- 1) The solution chosen can be wrong for the project.
- 2) The solution chosen can be **dependent on developer skills.**
- 3) The solution can generate a key person risk.
- 4) **Being a full stack developer is increasingly complex.**

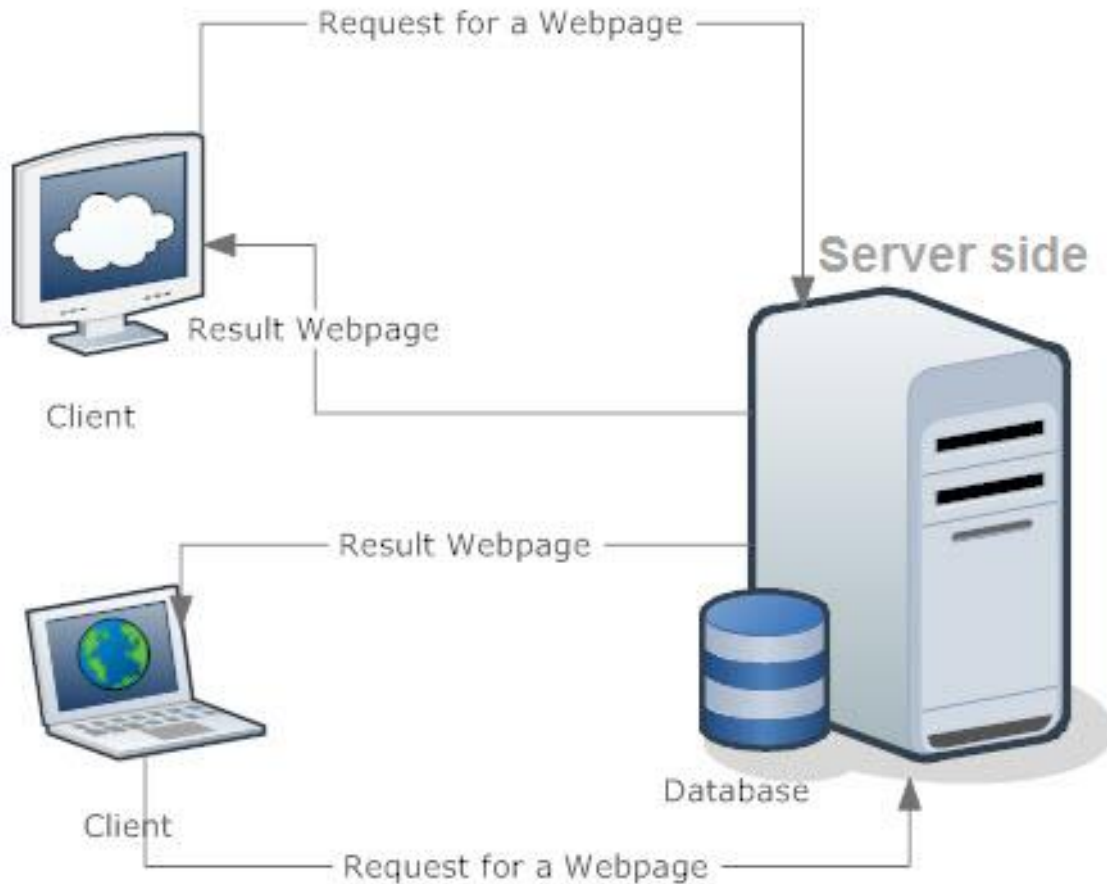
Diskusikan dengan kelompok anda, perbedaan antara DevOps dan **model pengembangan tradisional** (Waterfall Model, V-Model, Incremental Model, Spiral Model, ~~Big Bang Model~~, Prototype Model, RAD (Rapid Application Development) Model). Jelaskan pendapat anda di depan class.

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2 – What is JavaScript?

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Client Side Scripting



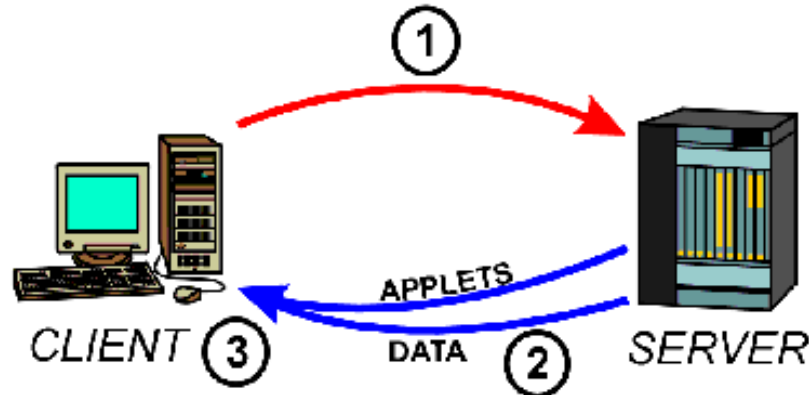
Client-side Scripting adalah bahasa pemrograman web yang pengolahan datanya dilakukan oleh komputer pengguna/pengunjung.

Jadi, ketika users berkunjung ke sebuah website, maka computer pengguna akan mendownload data/script yang bersifat client-side di web tersebut.

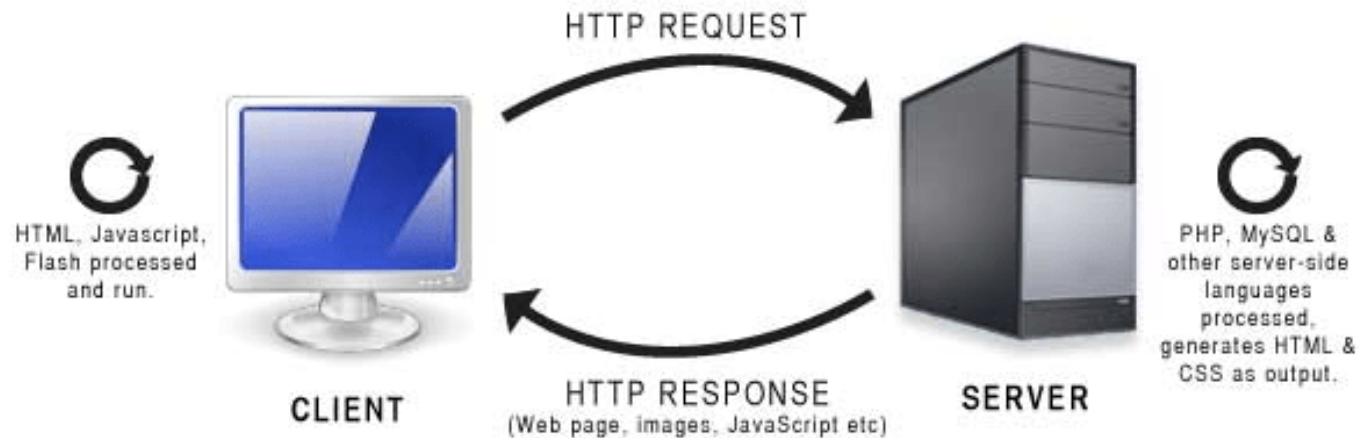
Contoh client side scripting adalah CSS, JavaScript, JQuery, HTML, XML dan HTML.

Contoh server side scripting adalah: JSP (Java Server Pages), ASP (Active Server Pages), PHP (Hypertext Preprocessor), Server Side Includes (SSI), Lasso, dan ColdFusion.

Client-side Configuration



1. Client sends request to server
2. Server processes request and returns information as needed
3. Data is processed by client's computer



Client Side *vs* Server Side



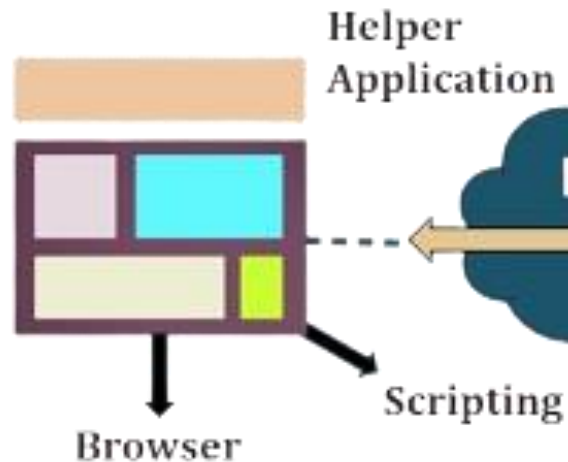
When a client (your computer) makes a request for a web page that information is processed by the web server. If the request is a server side script (e.g. Perl or PHP) before the information is returned to the client the script is executed on the server and the results of the script is returned to the client.



Once the client receives the returned information from the server if it contains a client side script (e.g. JavaScript) your computer browser executes that script before displaying the web page.

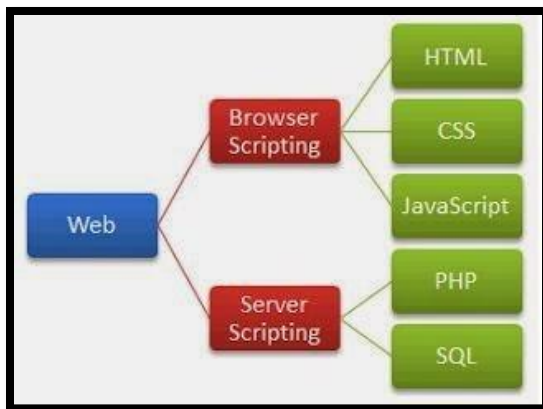
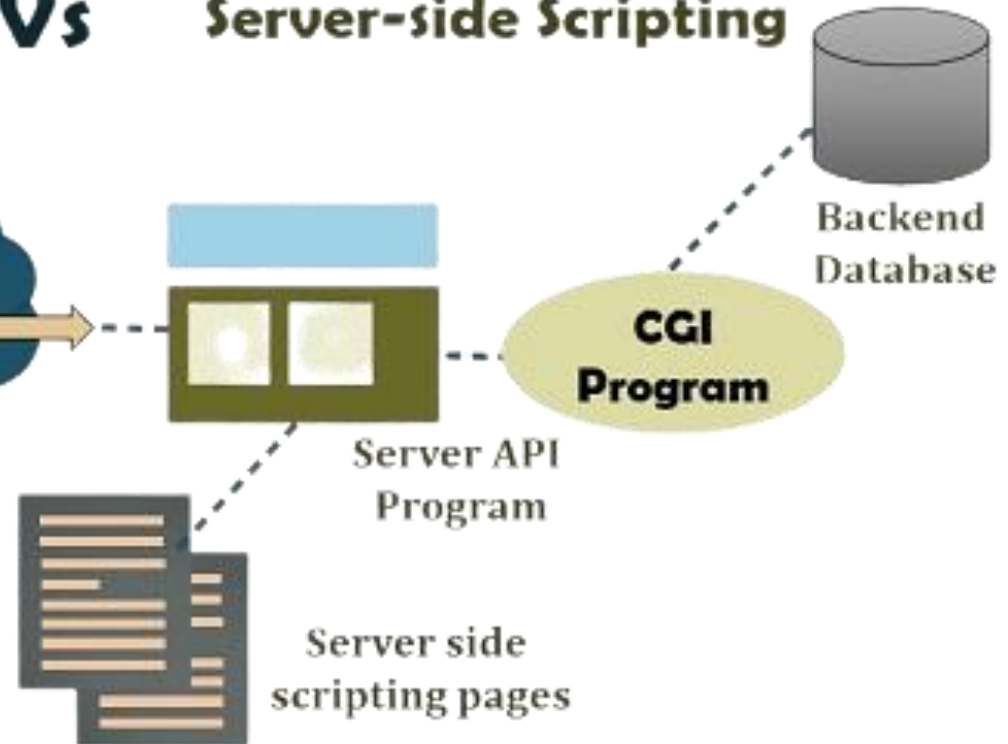
Client Side *vs* Server Side Scripts

Client-side Scripting



Vs

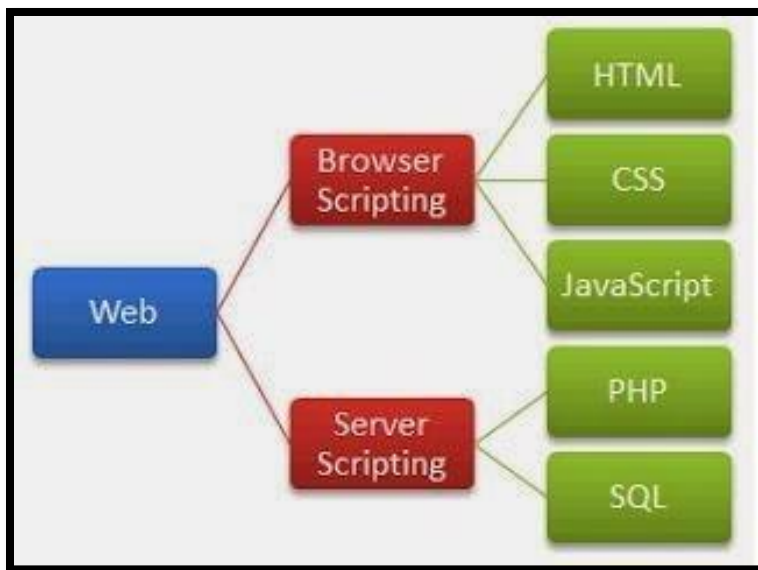
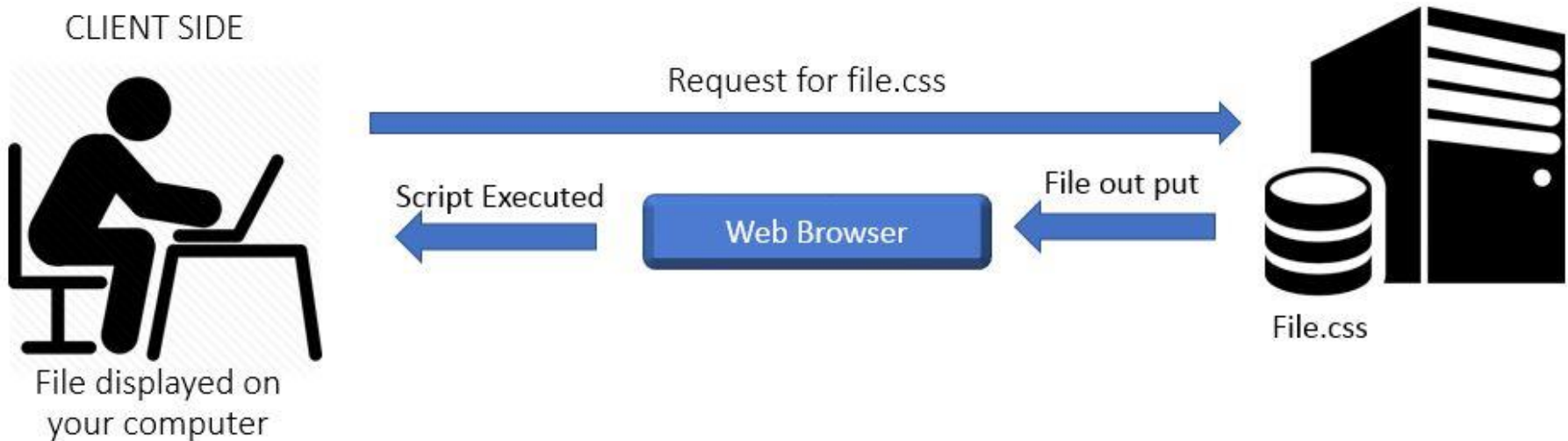
Server-side Scripting



What is CGI Web page?

The common gateway interface (CGI) is a standard way for a Web server to pass a Web user's request to an application program and to receive data back to forward to the *user*.

Process of Client Side Scripting



JavaScript vs Java

JAVA	Java Script
• Compiled • Mainly used for back-end • Executed in JVM or in the browser • Allows better security • Static type checking • The syntax is similar to C++ • Requires Java Development Kit (JDK) • For various apps	• Interpreted • Mainly used for front-end • Executed in the browser • Needs more effort to enhance security • Dynamic type checking • The syntax is similar to C • Can be written in any text editor • Mainly for web apps

- ❖ JavaScript and Java are completely different languages, both in concept and design.
- ❖ JavaScript was invented by Brendan Eich in 1995, and became an ECMA standard in 1997.
- ❖ ECMA-262 is the official name of the standard. ECMAScript is the official name of the language.
- ❖ A *compiler* translates the entire source code in a single run. An *interpreter* translates the entire source code line by line.



V/S



Interpreter mengkonversi *source code* menjadi *machine code* secara langsung ketika program dijalankan.



Pada compiler, *source code* akan dikonversi menjadi *machine code* sebelum program tersebut dijalankan.



JavaScript vs Java



JavaScript

- Server and Client Side Language
- Used for both UI and core business logic



Java

- Server side language
- Primarily used for core business logic

Feature of JavaScript

- ❖ **Scripting language and not Java:** In fact, JavaScript has nothing to do with Java. Then why is it called “Java” Script? When JavaScript was first released it was called Mocha, it was later renamed to LiveScript and then to JavaScript when Netscape (founded JavaScript) and Sun did a license agreement.
- ❖ **Object-based scripting language** which supports polymorphism, encapsulation and to some extent *inheritance* as well.
- ❖ **Interpreted language:** It doesn't have to be compiled like Java and C which require a compiler.
- ❖ **JavaScript runs in a browser:** We can run it on Google Chrome, Internet Explorer, Safari, etc. JavaScript can execute not only in the browser but also on the *server* and any device which has a JavaScript Engine.

TOP 10

Popular Programming Languages in 2020

1 Python

2 JavaScript

3 Java

4 C#

5 C

6 C++

7 GO

8 R

9 Swift

10 PHP

WWW.NORTHEASTERN.EDU/GRADUATE

What *can* we
do with
Javascript?

Why Javascript?

- 1) JavaScript Can Change HTML Content
- 2) JavaScript Can Change HTML Attribute Values
- 3) JavaScript Can Change HTML Styles (CSS)
- 4) JavaScript Can Hide HTML Elements
- 5) JavaScript Can Show HTML Elements

Companies who use JavaScript on:

Front-End

Facebook
Google
Quora
Uber

*99% of all
top websites*

Back-End

Walmart
LinkedIn
PayPal
Uber

many more...

Mobile Apps

Facebook
Instagram
Uber
Skype

many more...

Desktop Apps

Microsoft
VS Code
WhatsApp
Slack

many more...

JavaScript Capabilities

Comparing Programming Language Capabilities:

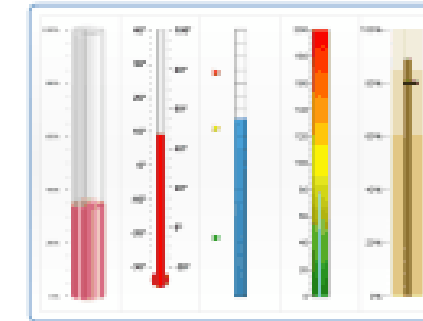
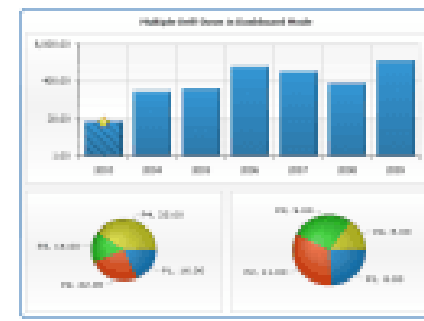
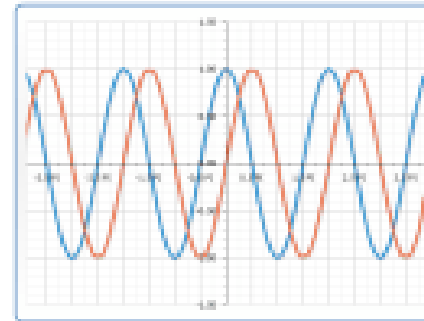
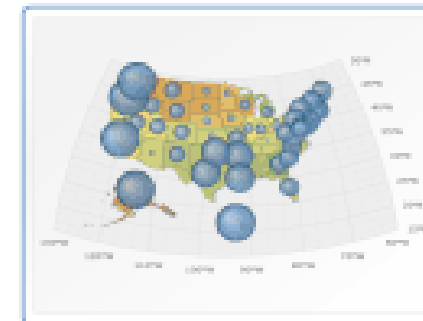
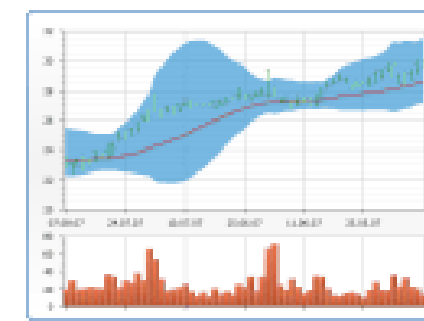
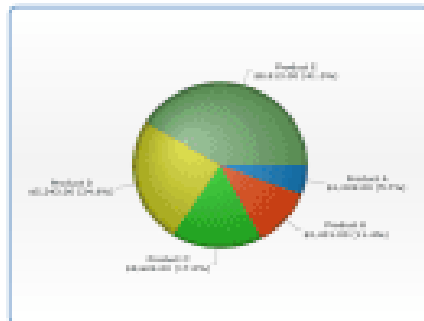
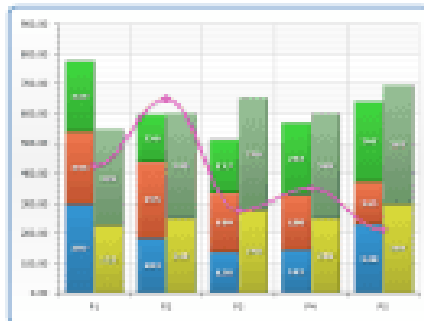
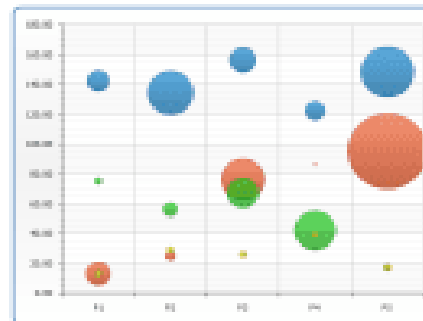
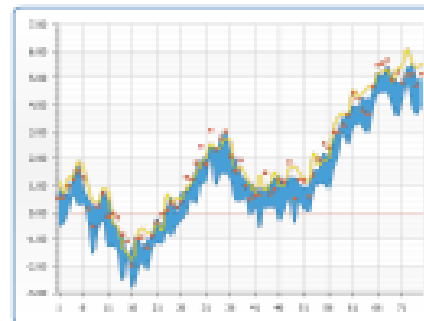
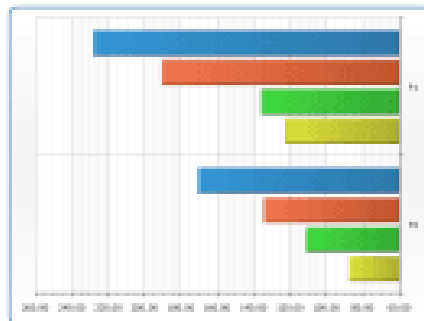
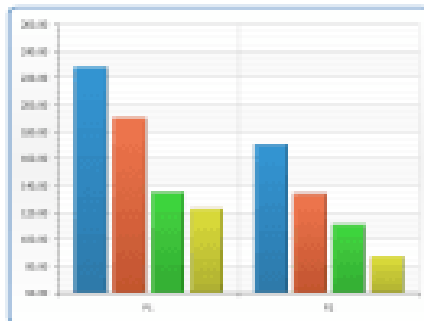
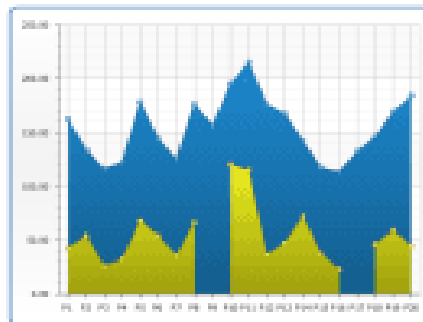
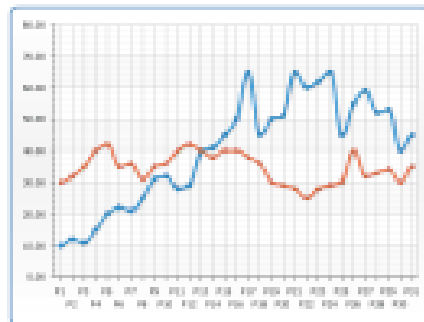
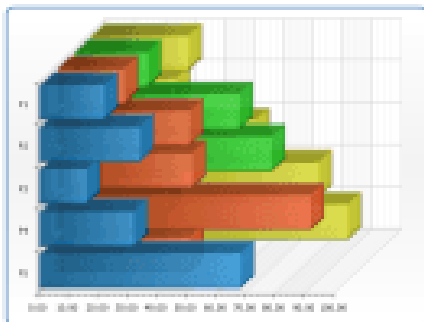
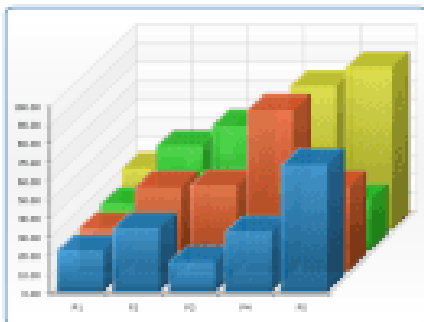
	JavaScript	C#	Java	Ruby	Python
Front-End	✓	✗	✗	✗	✗
Back-End	✓	✓	✓	✓	✓
Mobile Apps	✓	✓	✓	✓	✓
Desktop Apps	✓	✓	✓	✓	✓
Easy to learn	✓	✗	✗	✓	✓

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green with some visible veins and slight shadows, creating a natural border around the central text.

3 - JavaScript Graphics



Web programmers dream to create beautiful graphic designs in their webpage.



Can JavaScript *do* graphics?



- ✓ Typical uses for graphics include **banners**, ads from **sponsoring companies**, and colored **bullets** to *highlight important pieces of text*.
- ✓ The JavaScript scripting language can be used to enhance the graphics you put on the Web pages.
- ✓ **JavaScript can be used to dynamically control the graphic content of the page.**

How graphics *are* created in HTML?



The HTML `<canvas>` element can be used to draw graphics on a web page.

The graphic below is created with `<canvas>`:



It shows four elements: a red rectangle, a gradient rectangle, a multicolor rectangle, and a multicolor text.

Graphic Libraries

JavaScript libraries to use for both Artificial Intelligence graphs and other charts:

- ✓ Plotly.js
- ✓ Chart.js
- ✓ Google Chart

Plotly.js

Plotly.js is a charting library that comes with over 40 chart types, 3D charts, statistical graphs, and SVG maps.

Google Chart

From simple line charts to complex tree maps, Google Chart provides a number of built-in chart types:

- Scatter Chart
- Line Chart
- Bar / Column Chart
- Area Chart
- Pie Chart
- Donut Chart
- Org Chart
- Map / Geo Chart

Chart.js

Chart.js comes with many built-in chart types:

- Scatter
- Line
- Bar
- Radar
- Pie and Doughnut
- Polar Area
- Bubble

What is SVG?

- SVG stands for Scalable Vector Graphics
- SVG is used to define vector-based graphics for the Web
- SVG defines the graphics in XML format
- Every element and every attribute in SVG files can be animated
- SVG is a W3C recommendation
- SVG integrates with other W3C standards such as the DOM and XSL

HTML Canvas Graphics

What is HTML Canvas?

The HTML `<canvas>` element is used to draw graphics, on the fly, via JavaScript.

The `<canvas>` element is only a container for graphics. You must use JavaScript to actually draw the graphics.

Canvas has several methods for drawing paths, boxes, circles, text, and adding images.

Browser Support

The numbers in the table specify the first browser version that fully supports the `<canvas>` element.

Element					
<code><canvas></code>	4.0	9.0	2.0	3.1	9.0

Canvas Examples

A canvas is a rectangular area on an HTML page. By default, a canvas has no border and no content.

The markup looks like this:

```
<canvas id="myCanvas" width="200" height="100"></canvas>
```

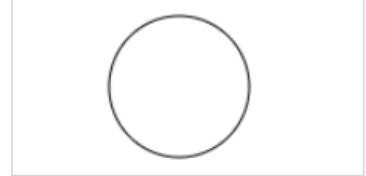
Draw a Circle

```
<!DOCTYPE html>
<html>
<body>

<canvas id="myCanvas" width="200" height="100" style="border:1px solid #d3d3d3;">
Your browser does not support the HTML canvas tag.</canvas>

<script>
  var c = document.getElementById("myCanvas");
  var ctx = c.getContext("2d");
  ctx.beginPath();
  ctx.arc(95,50,40,0,2*Math.PI);
  ctx.stroke();
</script>

</body>
</html>
```



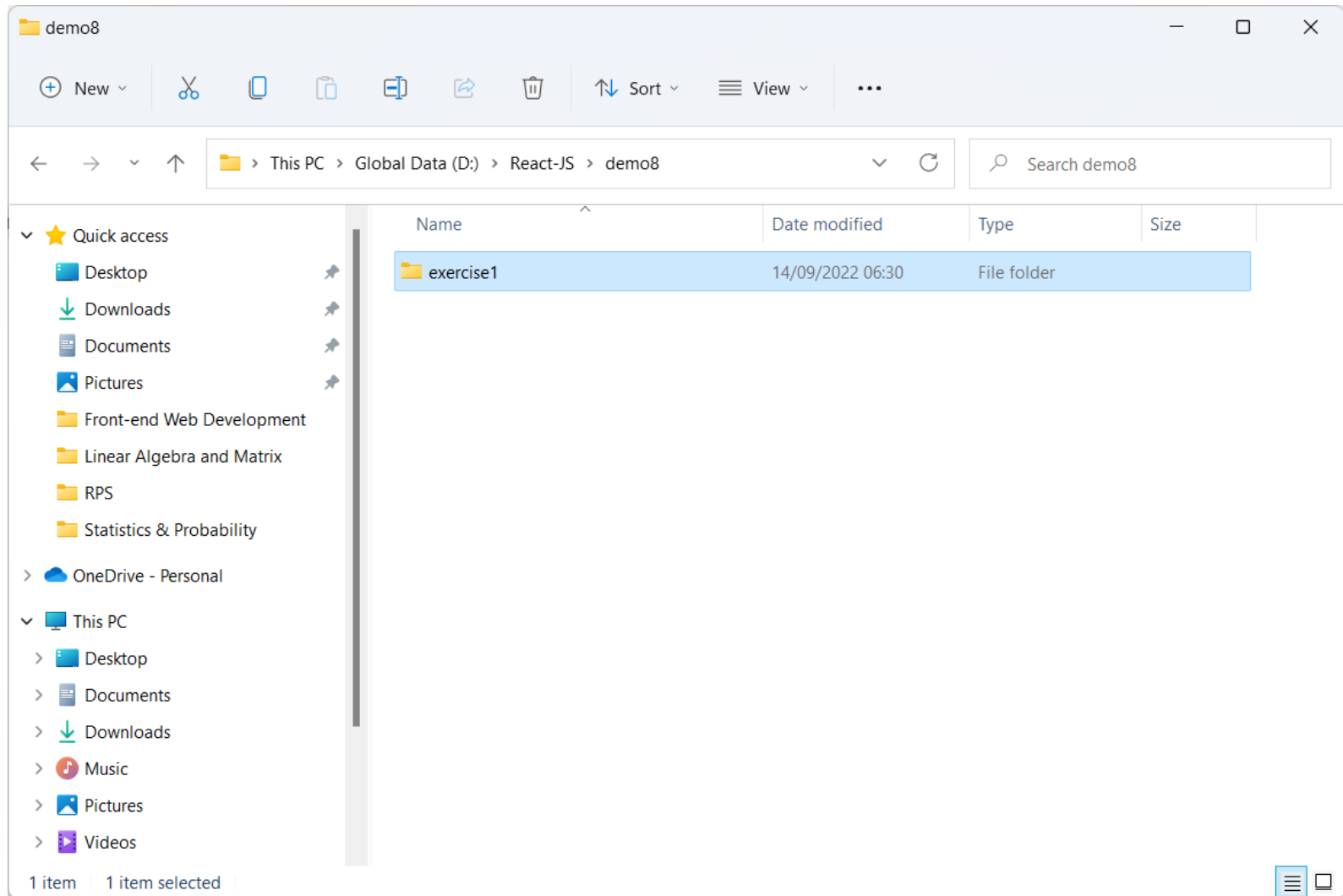
Canvas has APIs to draw circles and arcs:

arc(x, y, radius, startAngle, endAngle, anticlockwise): Draws an arc that is centered at (x, y) position with radius r starting at startAngle and ending at endAngle, going in the given direction indicated by anticlockwise (*defaulting to clockwise*).

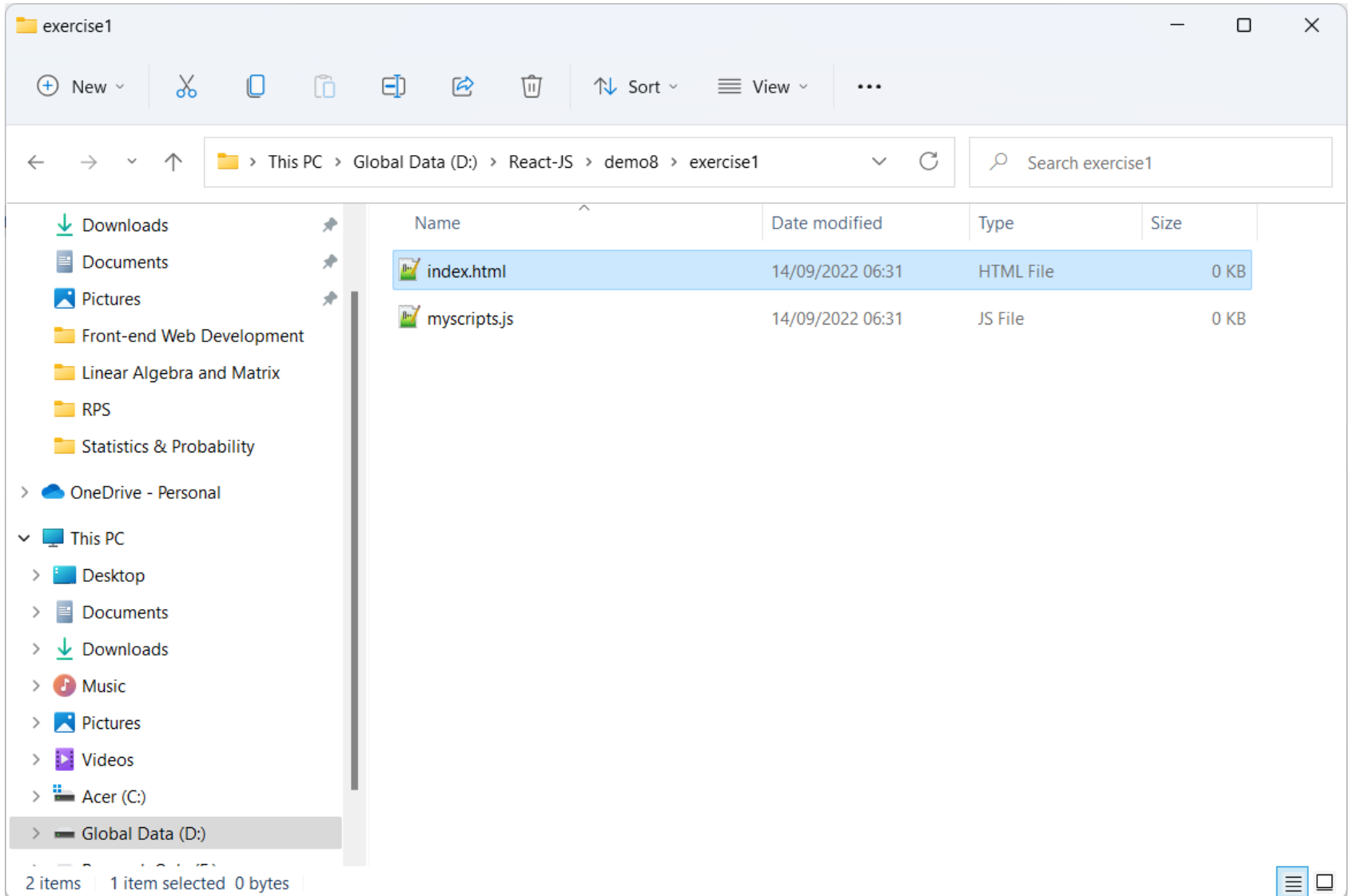


Exercise *for* Students

Create New Working Directory



Create 2 files for each exercise



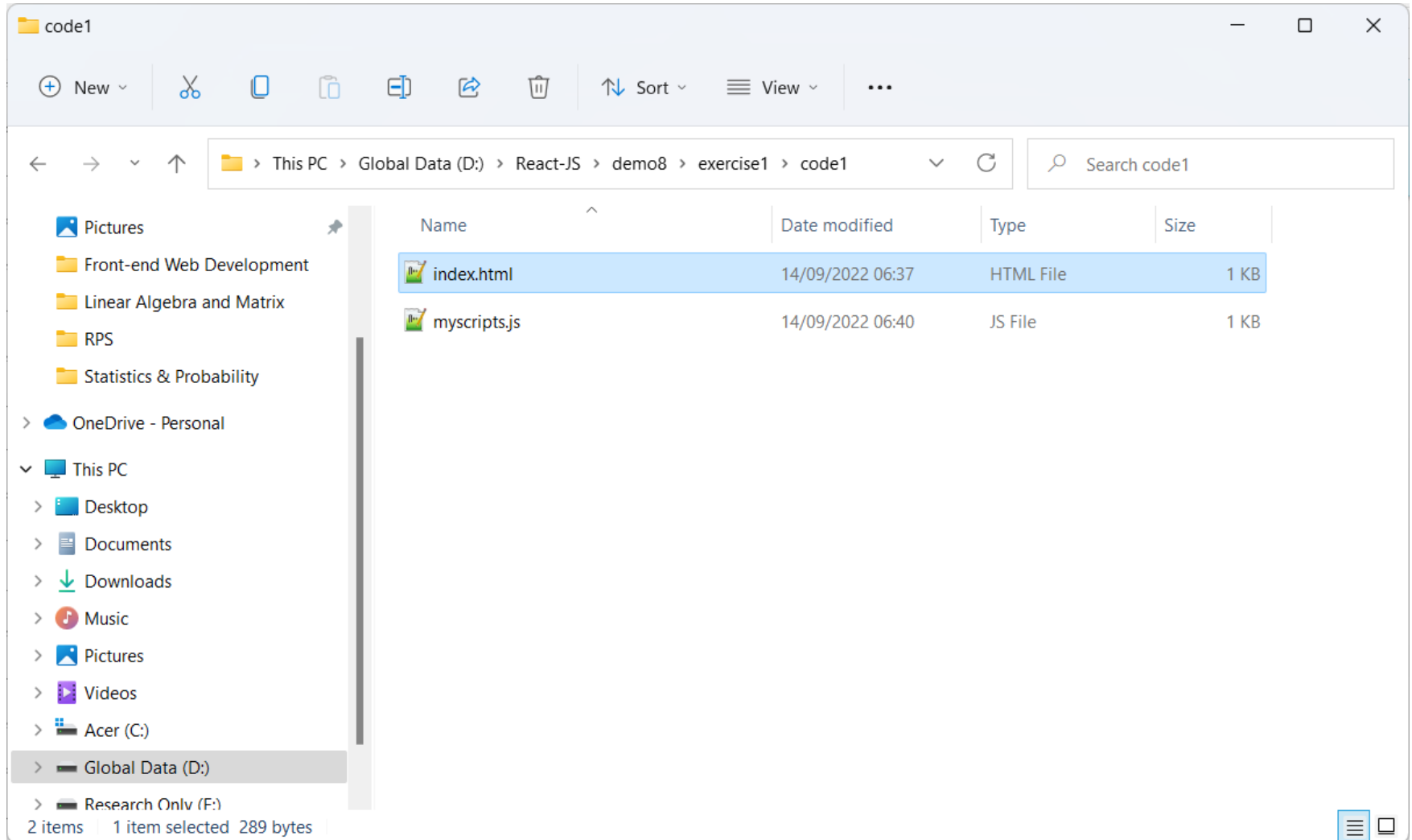
Exercise #1

Plotly.js Javascript Open Source Graphing Library

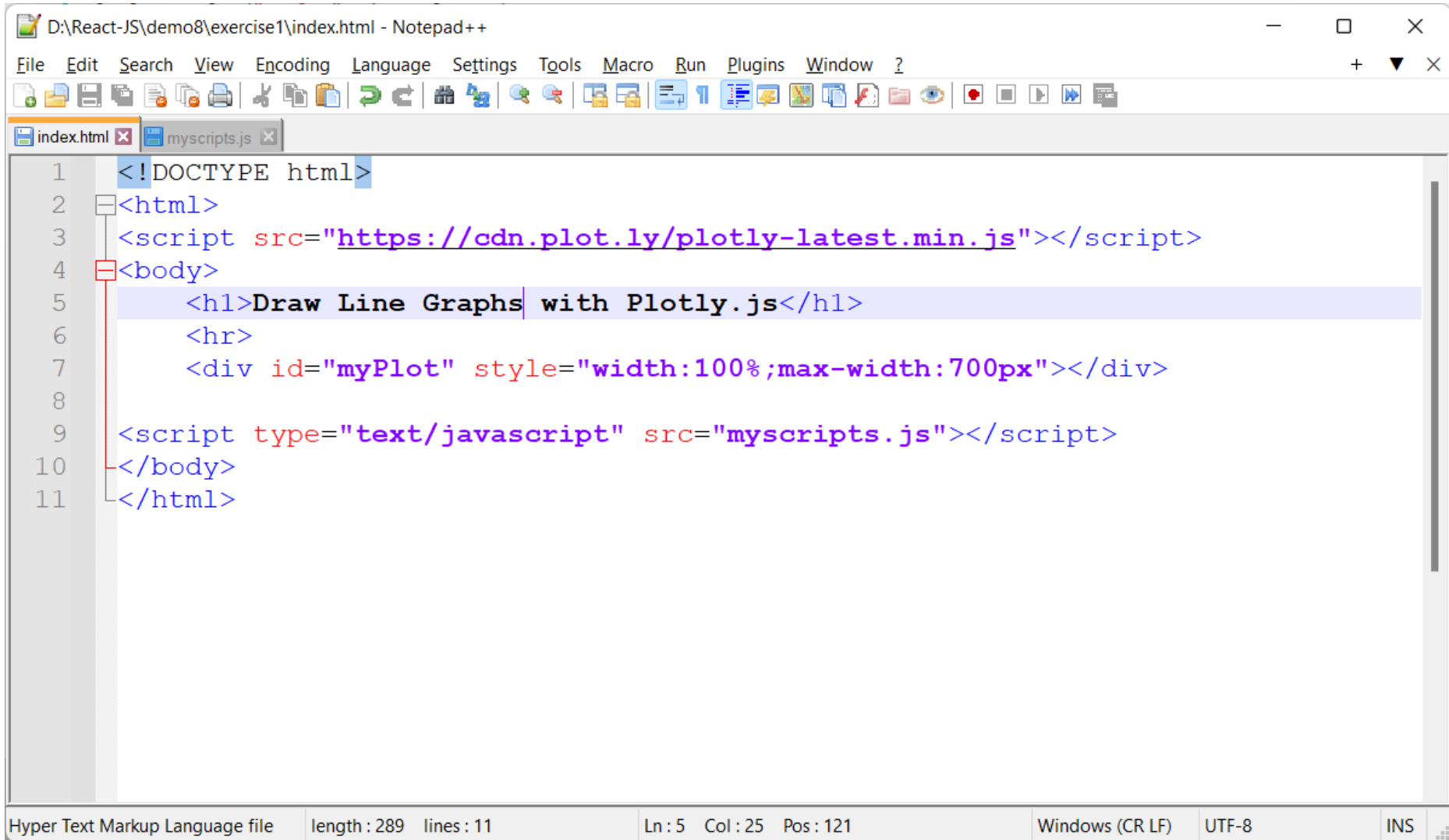
Code #1

(Line Graph)

Create HTML & JavaScript Files

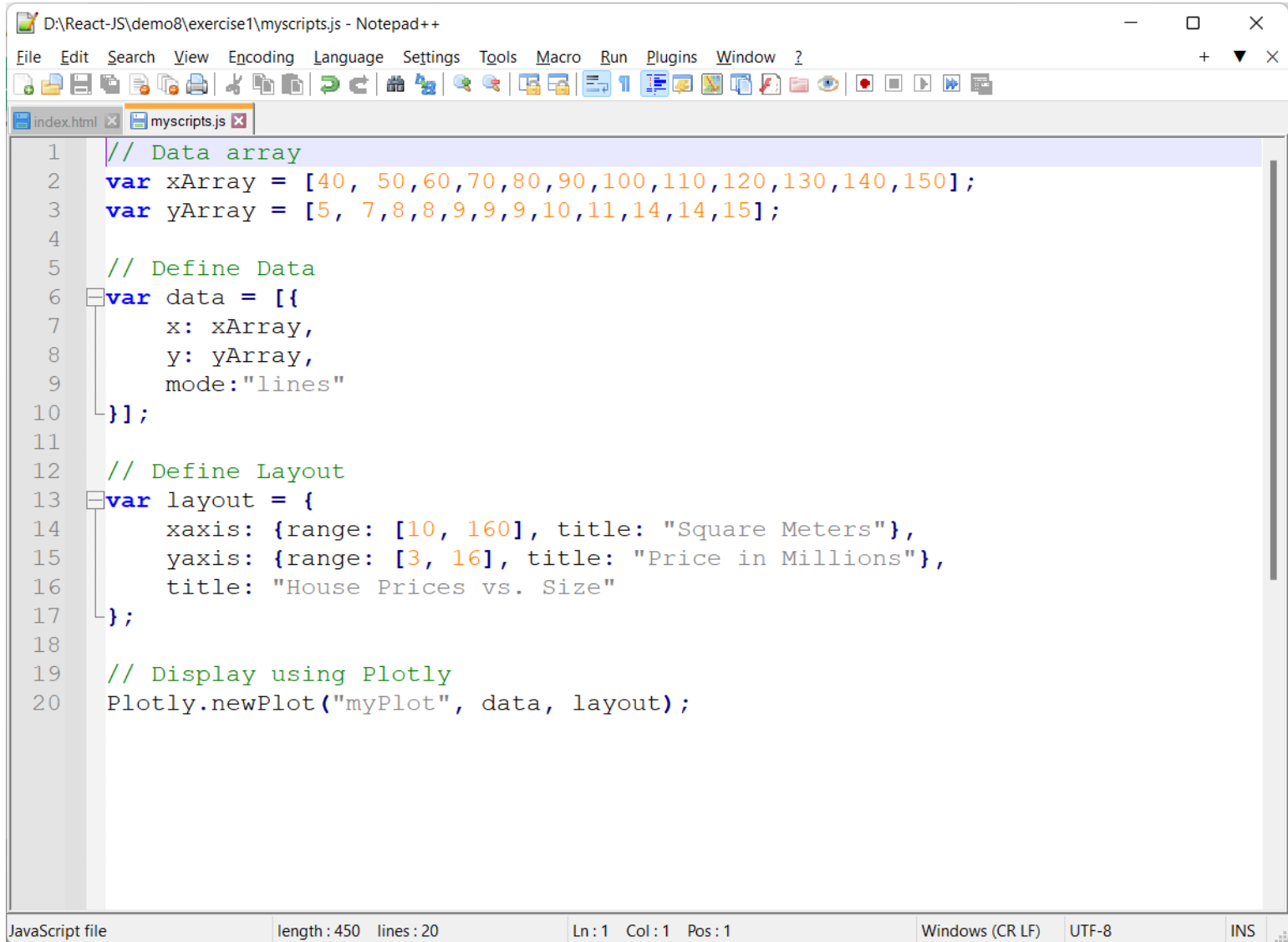


Write Code *in* HTML File



```
D:\React-JS\demo8\exercise1\index.html - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
index.html x myscripts.js x
1 <!DOCTYPE html>
2 <html>
3 <script src="https://cdn.plot.ly/plotly-latest.min.js"></script>
4 <body>
5     <h1>Draw Line Graphs with Plotly.js</h1>
6     <hr>
7     <div id="myPlot" style="width:100%;max-width:700px"></div>
8
9     <script type="text/javascript" src="myscripts.js"></script>
10 </body>
11 </html>
Hyper Text Markup Language file length : 289 lines : 11 Ln : 5 Col : 25 Pos : 121 Windows (CR LF) UTF-8 INS
```

Write Code *in* Javascript File

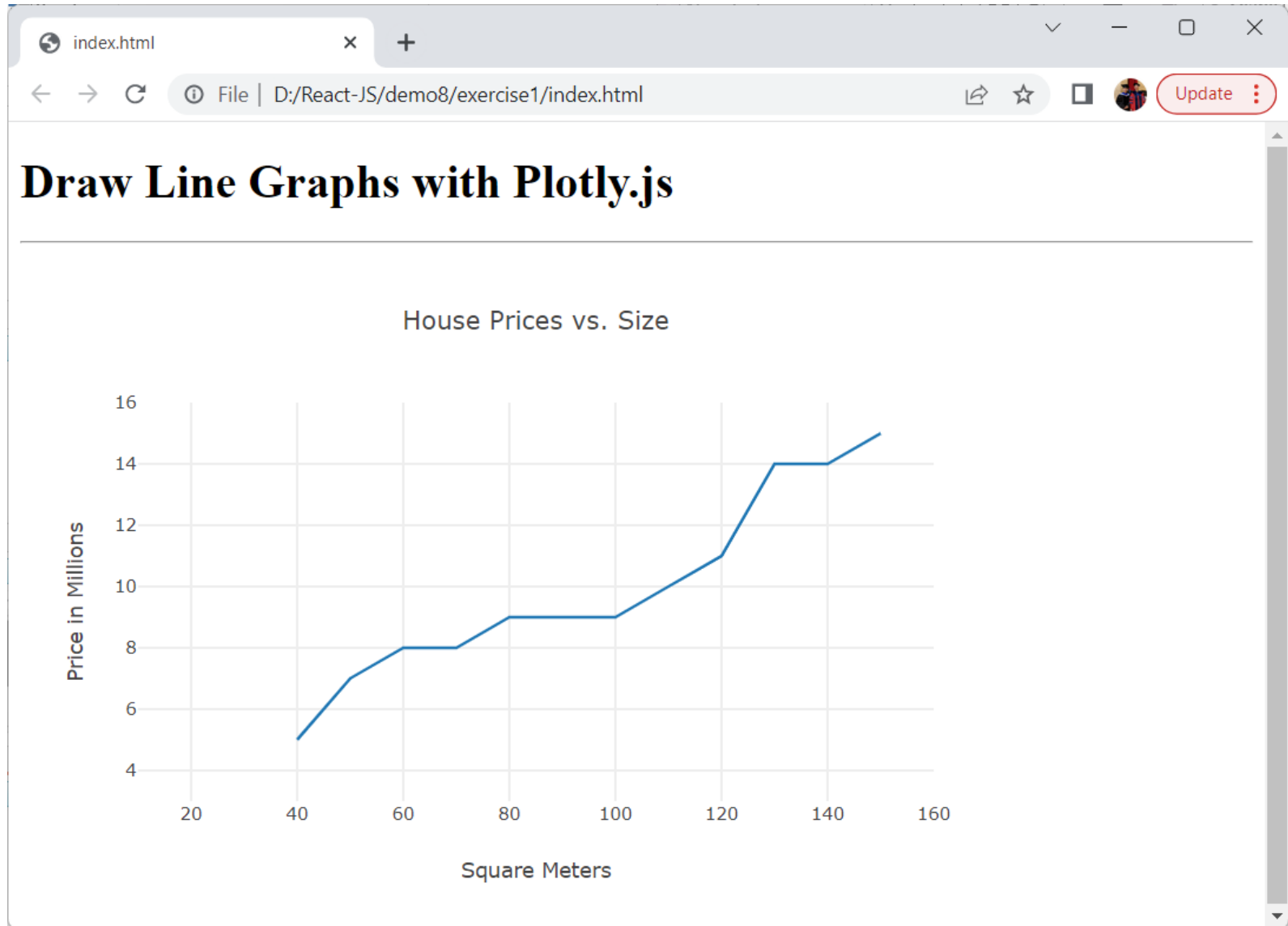


The screenshot shows a Notepad++ editor window with the title bar "D:\React-JS\demo8\exercise1\myscripts.js - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, Window, and Help. The toolbar contains various icons for file operations and editing. The editor has two tabs: "index.html" and "myscripts.js", with "myscripts.js" being the active tab. The code is as follows:

```
1 // Data array
2 var xArray = [40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 140, 150];
3 var yArray = [5, 7, 8, 8, 9, 9, 9, 10, 11, 14, 14, 15];
4
5 // Define Data
6 var data = [{
7     x: xArray,
8     y: yArray,
9     mode: "lines"
10 }];
11
12 // Define Layout
13 var layout = {
14     xaxis: {range: [10, 160], title: "Square Meters"},
15     yaxis: {range: [3, 16], title: "Price in Millions"},
16     title: "House Prices vs. Size"
17 };
18
19 // Display using Plotly
20 Plotly.newPlot("myPlot", data, layout);
```

The status bar at the bottom displays "JavaScript file", "length : 450 lines : 20", "Ln : 1 Col : 1 Pos : 1", "Windows (CR LF)", "UTF-8", and "INS".

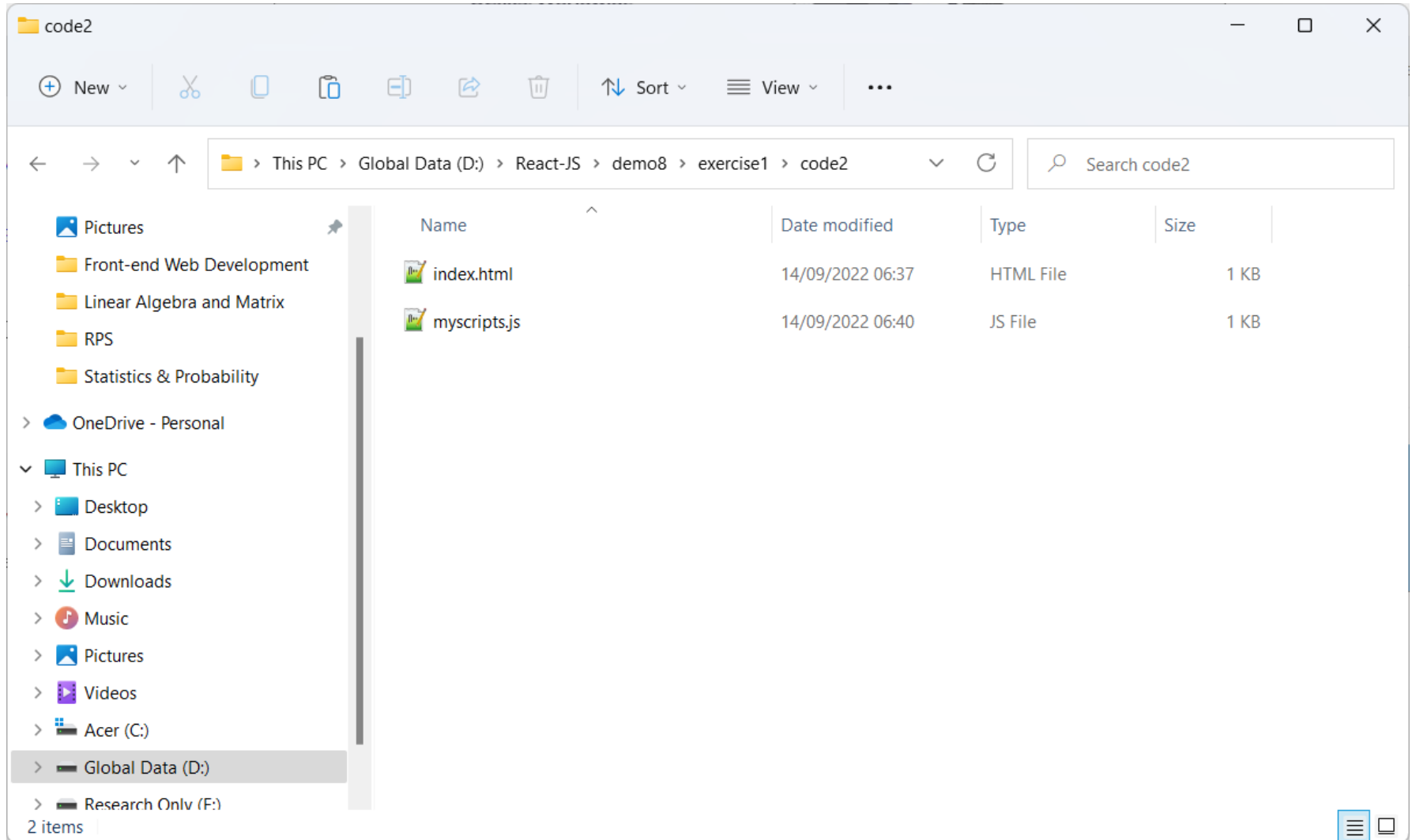
Expected Output



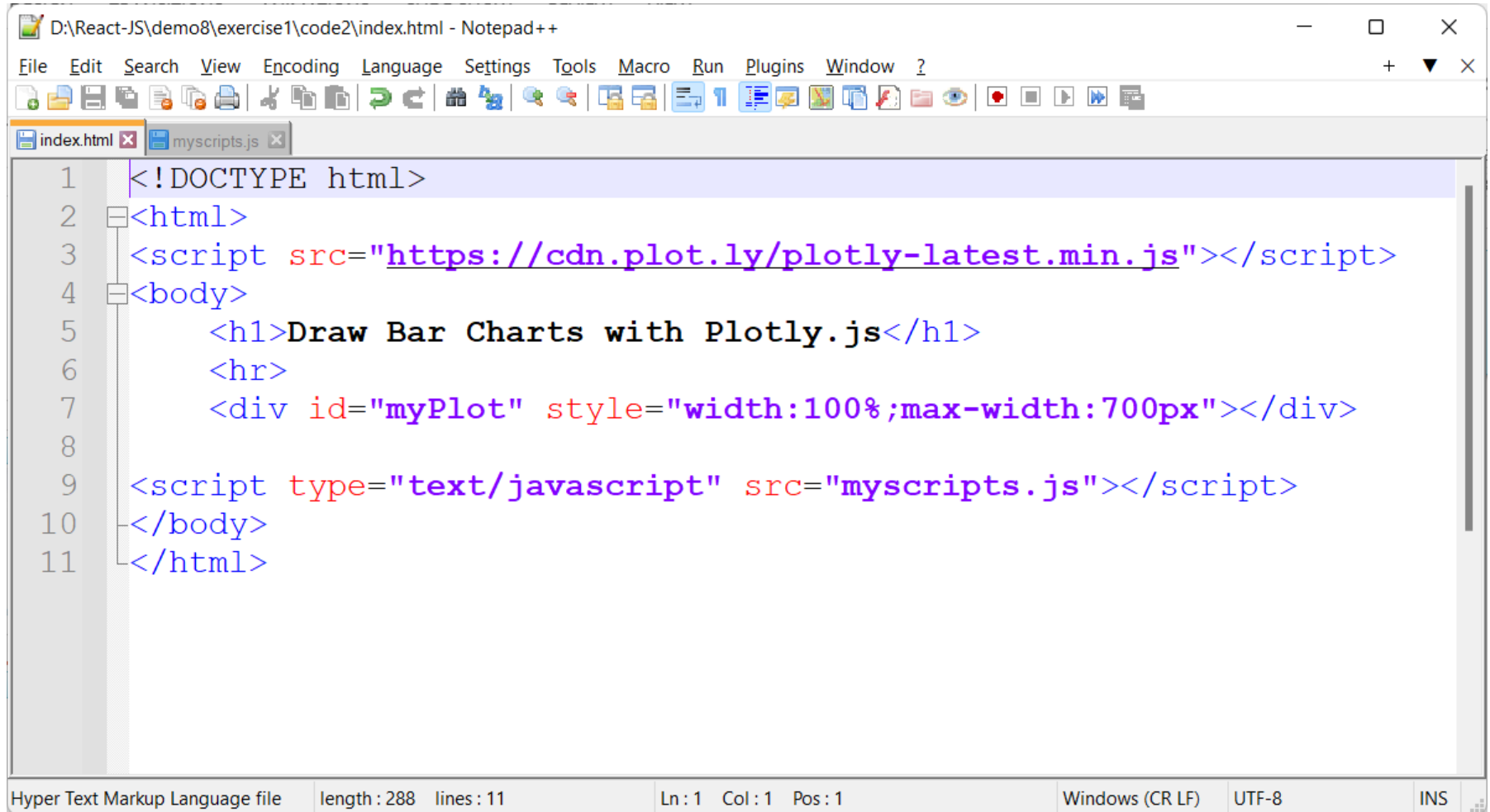
Code #2

(Bar Charts)

Create HTML & JavaScript Files



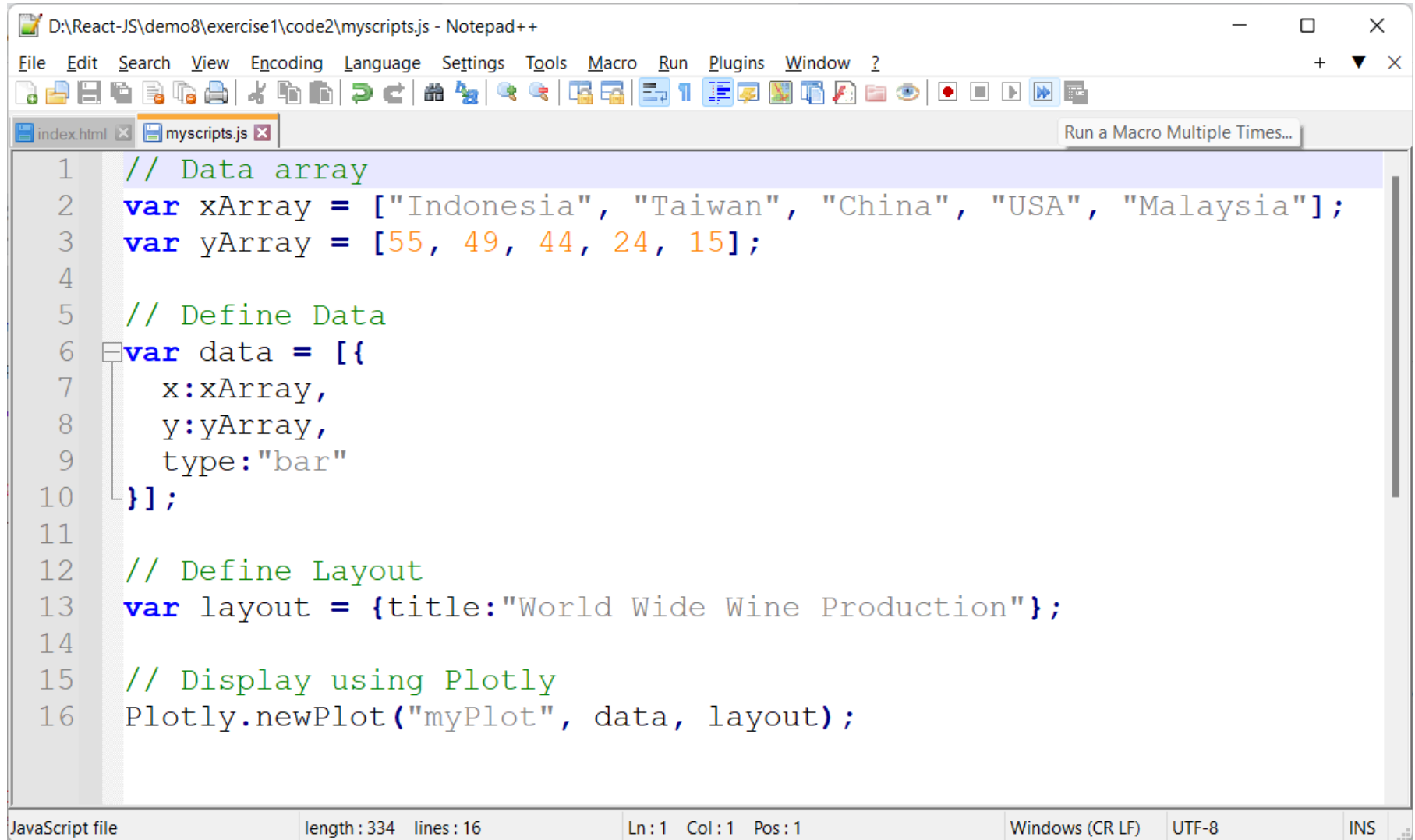
Write Code *in* HTML File



```
1 <!DOCTYPE html>
2 <html>
3   <script src="https://cdn.plot.ly/plotly-latest.min.js"></script>
4   <body>
5     <h1>Draw Bar Charts with Plotly.js</h1>
6     <hr>
7     <div id="myPlot" style="width:100%;max-width:700px"></div>
8
9     <script type="text/javascript" src="myscripts.js"></script>
10  </body>
11 </html>
```

Hyper Text Markup Language file length : 288 lines : 11 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS

Write Code *in* Javascript File



The image shows a Notepad++ editor window with the file path `D:\React-JS\demo8\exercise1\code2\myscripts.js`. The editor contains 16 lines of JavaScript code. Lines 1-4 define two arrays: `xArray` with country names and `yArray` with numerical values. Lines 5-10 define a `data` object with `x`, `y`, and `type` properties. Lines 11-13 define a `layout` object with a `title` property. Lines 14-16 use `Plotly.newPlot` to render the chart. The status bar at the bottom indicates the file is a JavaScript file, 334 characters long, 16 lines, and shows cursor position (Ln: 1, Col: 1, Pos: 1).

```
1 // Data array
2 var xArray = ["Indonesia", "Taiwan", "China", "USA", "Malaysia"];
3 var yArray = [55, 49, 44, 24, 15];
4
5 // Define Data
6 var data = [{
7   x:xArray,
8   y:yArray,
9   type:"bar"
10 }];
11
12 // Define Layout
13 var layout = {title:"World Wide Wine Production"};
14
15 // Display using Plotly
16 Plotly.newPlot("myPlot", data, layout);
```

JavaScript file length : 334 lines : 16 Ln: 1 Col: 1 Pos: 1 Windows (CR LF) UTF-8 INS

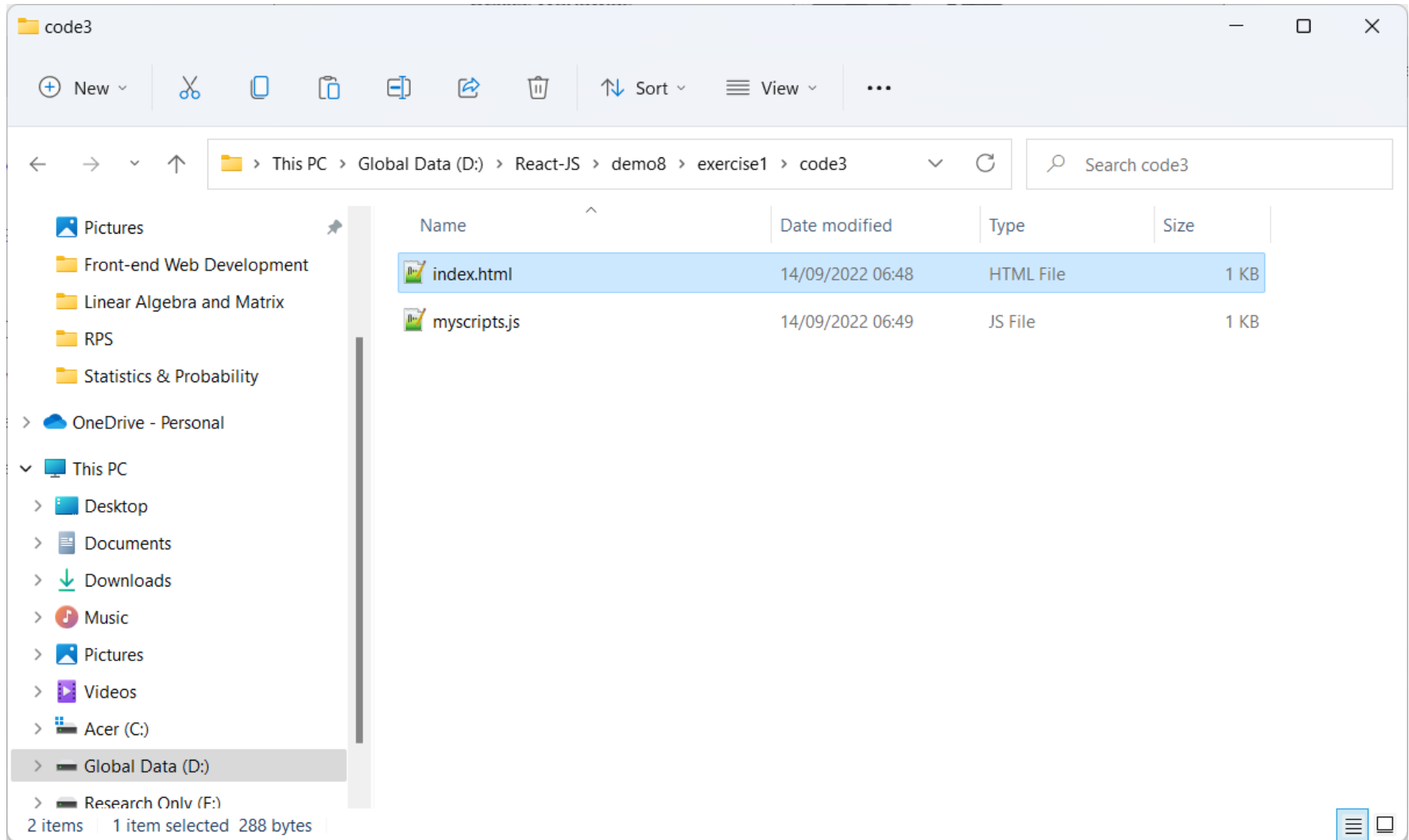
Expected Output



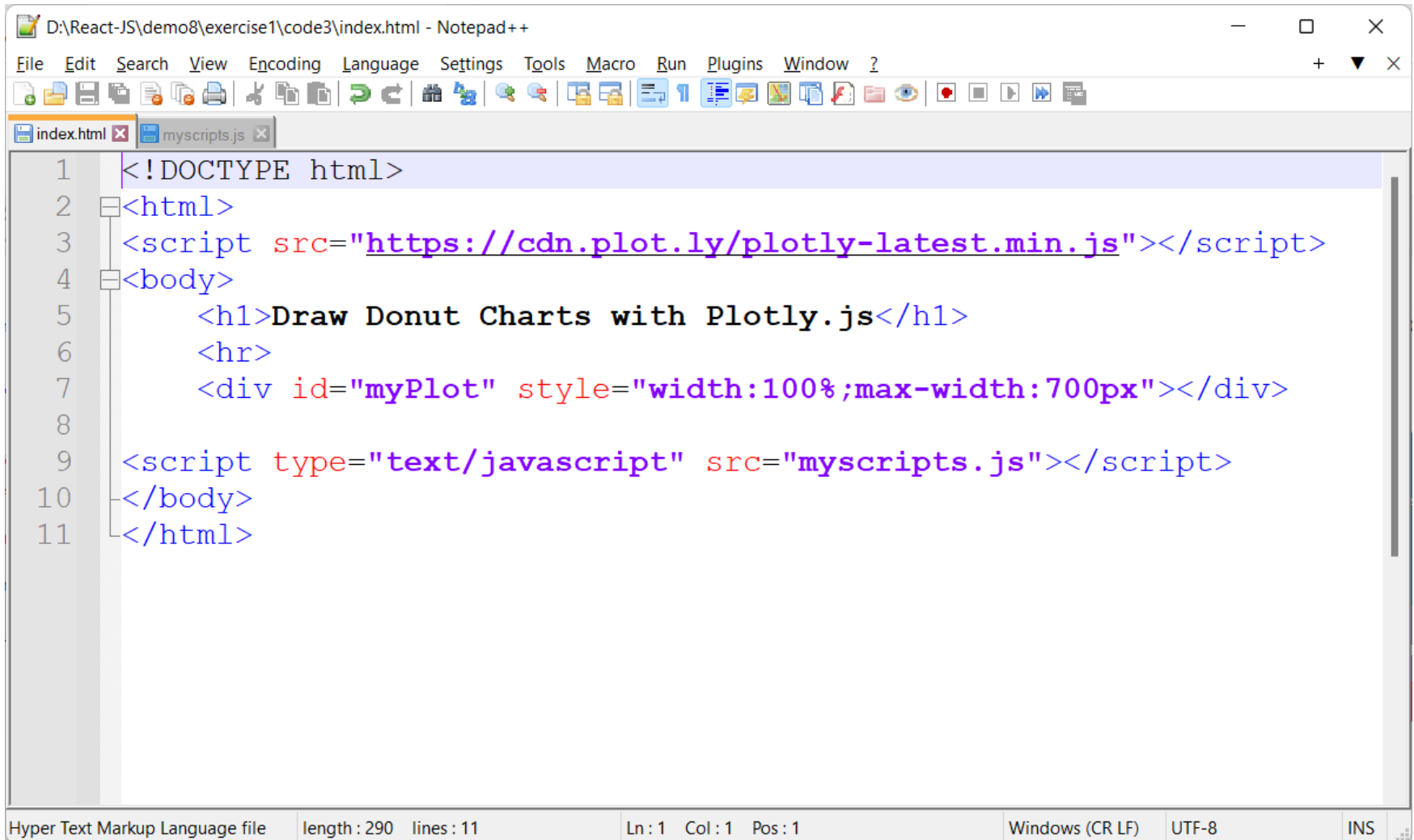
Code #3

(Donut Charts)

Create HTML & JavaScript Files

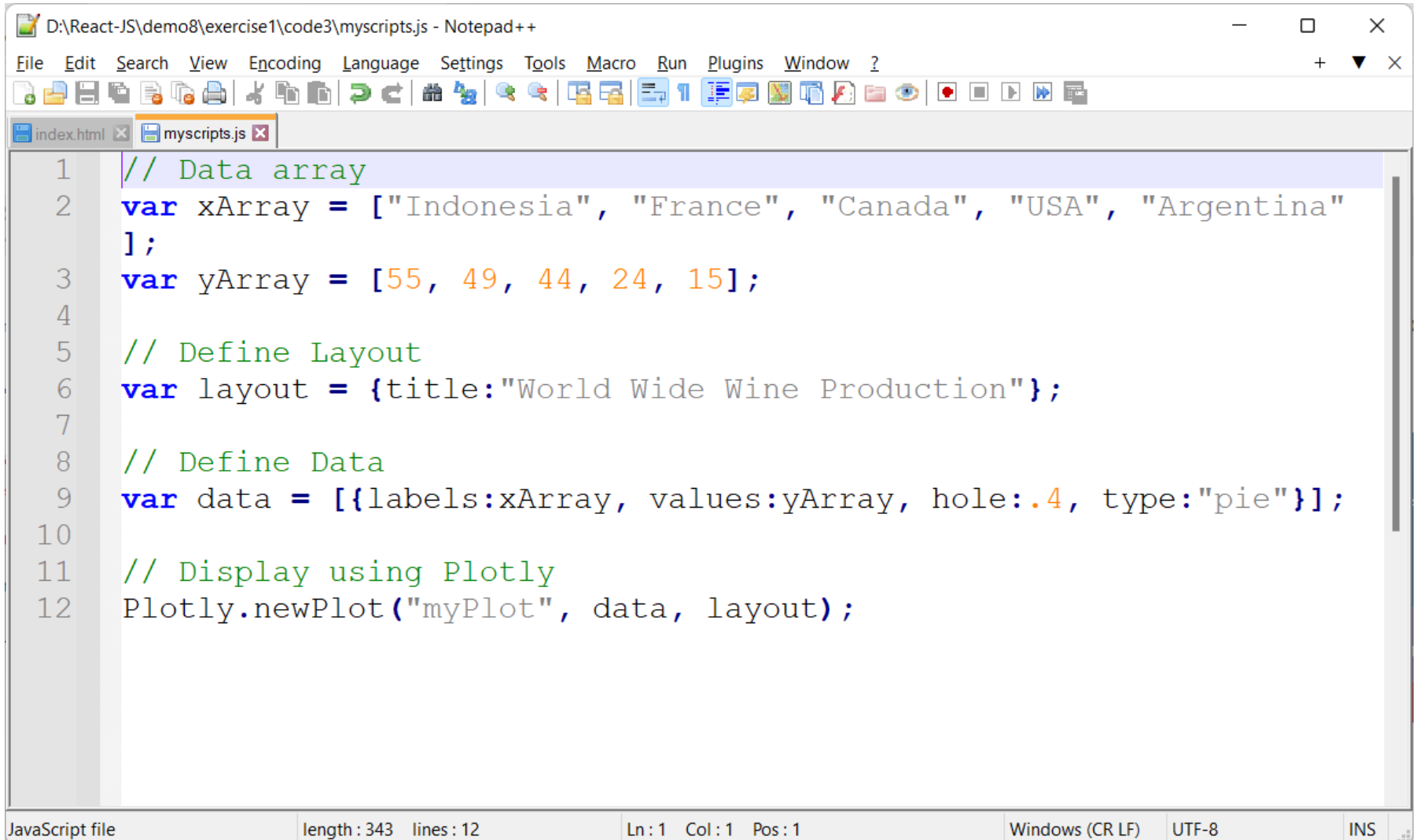


Write Code *in* HTML File



```
D:\React-JS\demo8\exercise1\code3\index.html - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
index.html x myscripts.js x
1 <!DOCTYPE html>
2 <html>
3 <script src="https://cdn.plot.ly/plotly-latest.min.js"></script>
4 <body>
5 <h1>Draw Donut Charts with Plotly.js</h1>
6 <hr>
7 <div id="myPlot" style="width:100%;max-width:700px"></div>
8
9 <script type="text/javascript" src="myscripts.js"></script>
10 </body>
11 </html>
Hyper Text Markup Language file length : 290 lines : 11 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS
```


Write Code *in* Javascript File

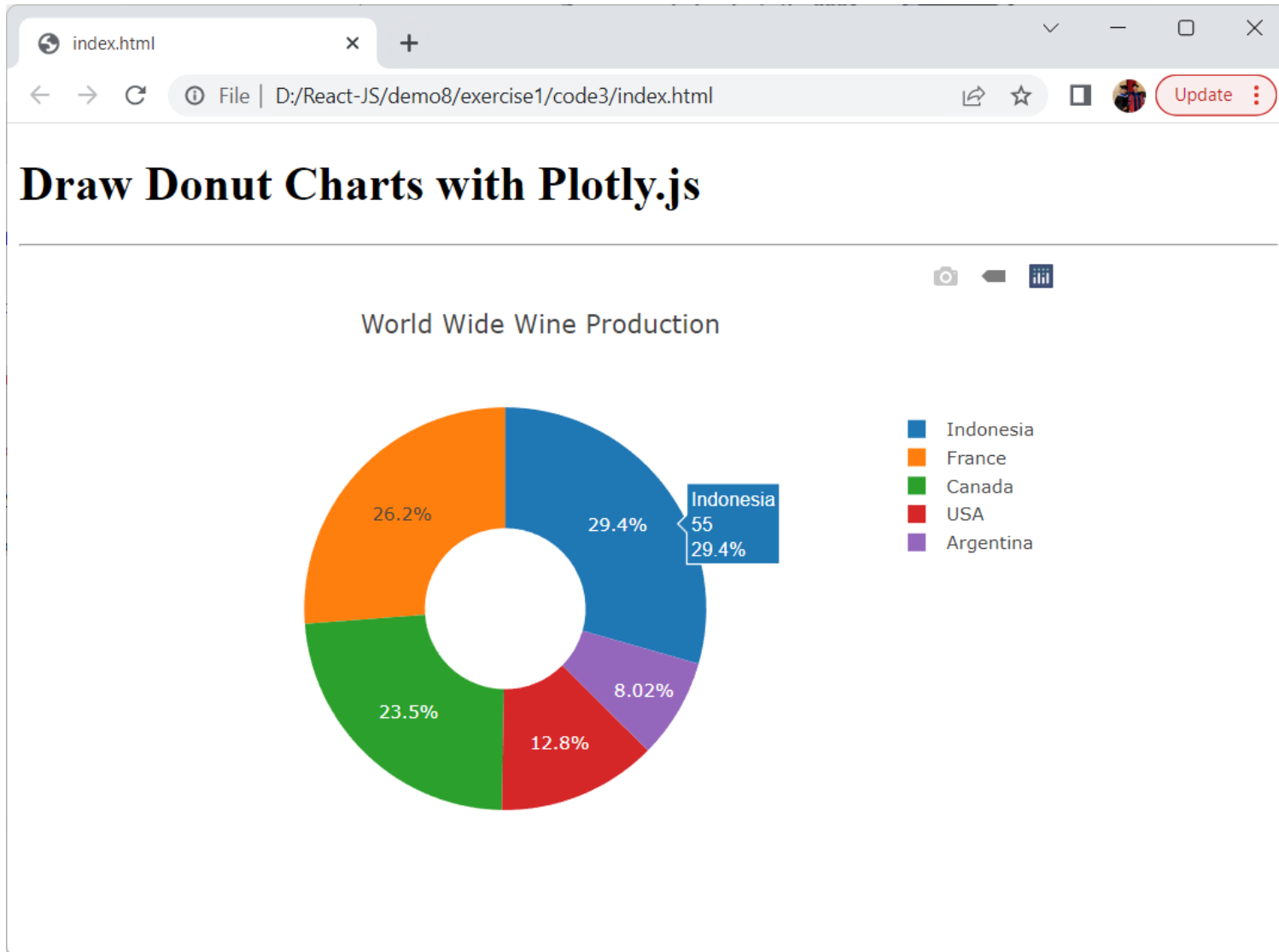


The image shows a Notepad++ window with the title bar "D:\React-JS\demo8\exercise1\code3\myscripts.js - Notepad++". The menu bar includes File, Edit, Search, View, Encoding, Language, Settings, Tools, Macro, Run, Plugins, and Window. The toolbar contains various icons for file operations and development. The editor has two tabs: "index.html" and "myscripts.js". The code in "myscripts.js" is as follows:

```
1 // Data array
2 var xArray = ["Indonesia", "France", "Canada", "USA", "Argentina"];
3 var yArray = [55, 49, 44, 24, 15];
4
5 // Define Layout
6 var layout = {title:"World Wide Wine Production"};
7
8 // Define Data
9 var data = [{labels:xArray, values:yArray, hole:.4, type:"pie"}];
10
11 // Display using Plotly
12 Plotly.newPlot("myPlot", data, layout);
```

The status bar at the bottom displays: "JavaScript file", "length : 343 lines : 12", "Ln : 1 Col : 1 Pos : 1", "Windows (CR LF)", "UTF-8", and "INS".

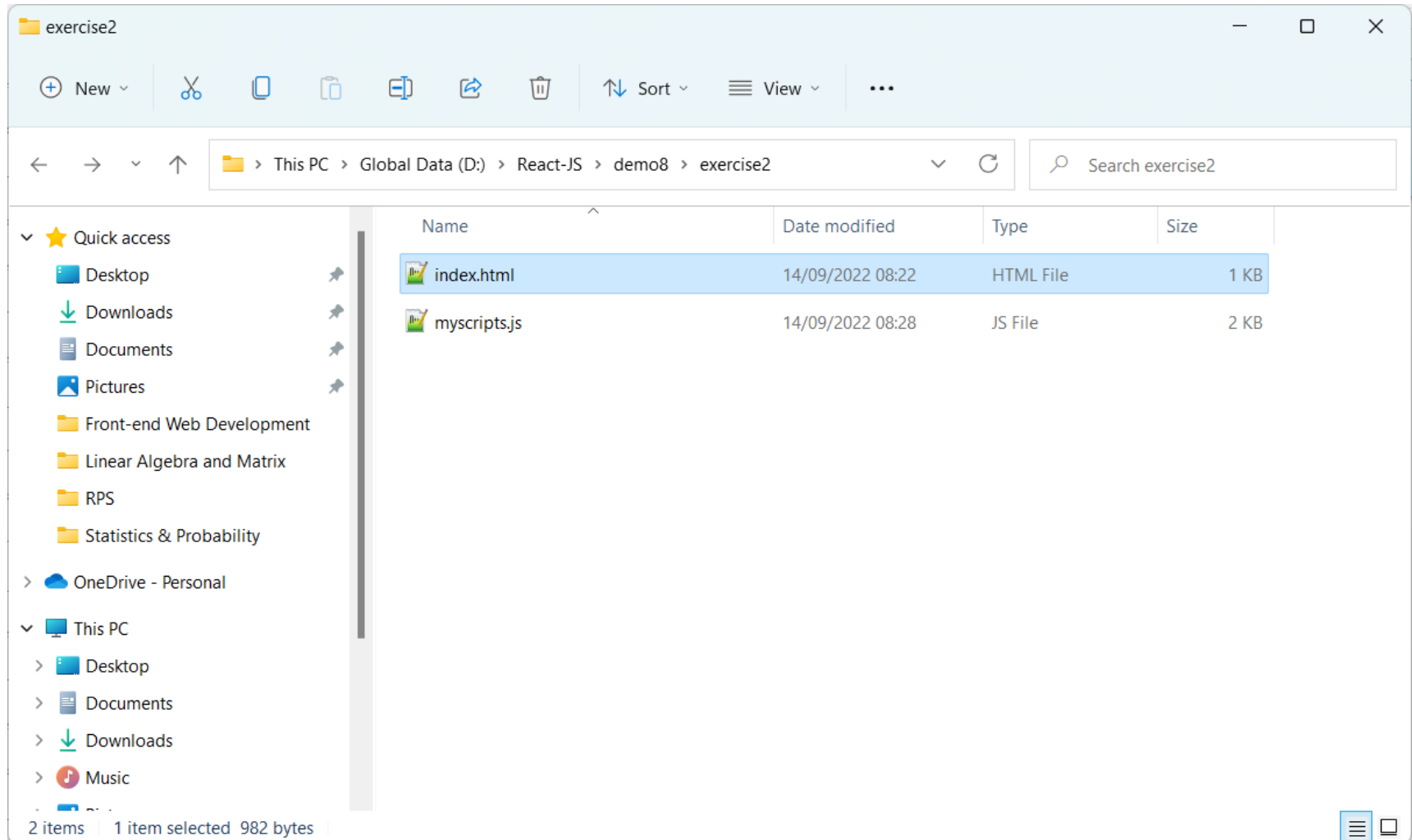
Expected Output



Exercise #2

Draw Dynamic Donut Charts with Plotly.js

Create HTML & JavaScript Files



Write Code *in* HTML File

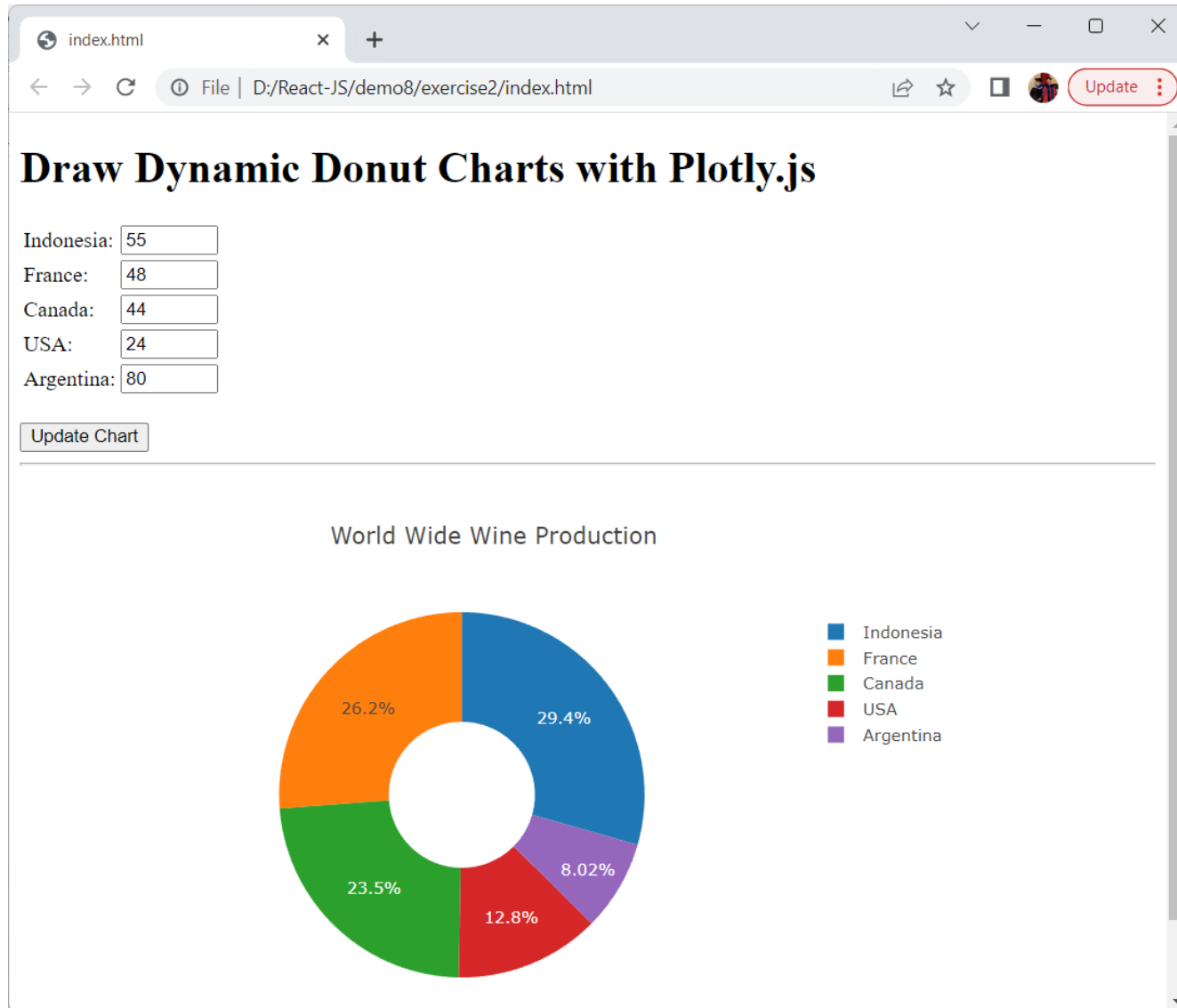
```
D:\React-JS\demo8\exercise2\index.html - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
index.html x myscripts.js x
1 <!DOCTYPE html>
2 <html>
3 <script src="https://cdn.plot.ly/plotly-latest.min.js"></script>
4 <body>
5   <h1>Draw Dynamic Donut Charts with Plotly.js</h1>
6
7   <table>
8     <tr>
9       <td><label>Indonesia:</label></td>
10      <td><input type="number" id="id-indo" min="10" max="200"></td>
11    </tr>
12    <tr>
13      <td><label>France:</label></td>
14      <td><input type="number" id="id-fran" min="10" max="200"></td>
15    </tr>
16    <tr>
17      <td><label>Canada:</label></td>
18      <td><input type="number" id="id-cana" min="10" max="200"></td>
19    </tr>
20    <tr>
21      <td><label>USA:</label></td>
22      <td><input type="number" id="id-usa" min="10" max="200"></td>
23    </tr>
24    <tr>
25      <td><label>Argentina:</label></td>
26      <td><input type="number" id="id-arge" min="10" max="200"></td>
27    </tr>
28  </table>
29  <br><button id="btn-update">Update Chart</button><br>
30  <hr>
31  <div id="myPlot" style="width:100%;max-width:700px"></div>
32
33  <script type="text/javascript" src="myscripts.js"></script>
34 </body>
35 </html>
Hyper Text Markup Language file length : 982 lines : 35 Ln : 1 Col : 1 Pos : 1 Windows (CR LF) UTF-8 INS
```

Write Code *in* Javascript File

```
D:\React-JS\demo8\exercise2\myscripts.js - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ? + ▼ X

index.html x myscripts.js x
1 // create fuction to plot chart.
2 function plotDonut(indo, france, canada, usa, argen){
3     // Data array
4     var xArray = ["Indonesia", "France", "Canada", "USA", "Argentina"];
5     var yArray = [indo, france, canada, usa, argen];
6
7     // Define Layout
8     var layout = {title:"World Wide Wine Production"};
9
10    // Define Data
11    var data = [{labels:xArray, values:yArray, hole:.4, type:"pie"}];
12
13    // Display using Plotly
14    Plotly.newPlot("myPlot", data, layout);
15 }
16
17 // view default donut chart
18 plotDonut(55, 49, 44, 24, 15);
19
20 // Set default values to input fields
21 document.getElementById("id-indo").value = "55";
22 document.getElementById("id-fran").value = "48";
23 document.getElementById("id-cana").value = "44";
24 document.getElementById("id-usa").value = "24";
25 document.getElementById("id-arge").value = "80";
26
27 // Get the button, and when the user clicks on it, execute myFunction
28 document.getElementById("btn-update").onclick = function() {
29     // get input values
30     var val_indo = document.getElementById("id-indo").value;
31     var val_fran = document.getElementById("id-fran").value;
32     var val_cana = document.getElementById("id-cana").value;
33     var val_usa = document.getElementById("id-usa").value;
34     var val_arge = document.getElementById("id-arge").value;
35
36     // Update donut chart
37     plotDonut(val_indo, val_fran, val_cana, val_usa, val_arge);
38 };
39
40
41
length: 1.360 lines: 41 Ln: 41 Col: 1 Pos: 1.361 Windows (CR LF) UTF-8 INS
```

Expected Output



Test your understanding

The following are some questions to reinforce your understanding of the concepts we've covered so far.



Question #1

Jelaskan dengan singkat perbedaan Front-end dan Back-end web developer?
Selanjutnya, jelaskan tanggung jawab dari seorang DevOps Engineer?

Question #2

Jelaskan dengan singkat kelebihan dan kekurangan jika menjadi seorang Full-stack web developer?

Question #3

Jelaskan apa yang dimaksud dengan istilah “Modern Web Development”???

Question #4

Apa peran seorang DevOps Engineer dalam pengembangan dan pengelolaan aplikasi /software/perangkat lunak?

Question #5

Bagaimana seorang DevOps Engineer memastikan keamanan dan ketersediaan aplikasi web yang mereka kelola?

END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

